

Haematopoietic Cell Transplantation and Cellular Therapies

Study-group handout · chapter by chapter
Outlines · tables · flowcharts

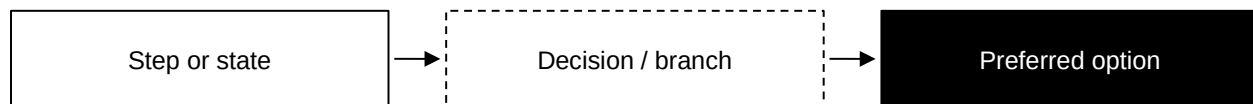
95 chapters · Parts I–IX

Source: Sureda A, Corbacioglu S, Greco R, Kröger N, Carreras E (eds). The EBMT Handbook, 8th ed. Springer 2024. Open access, CC BY 4.0.

How to read these slides

- One slide per handbook chapter, in book order. Header shows Part and chapter number.
- Three forms only: **outline** (what to know), **table** (compare / thresholds), **flowchart** (sequence or decision).
- Footer marks provenance: ● content checked against the handbook text; ○ condensed summary — confirm details in the chapter before clinical use.
- Doses and cut-offs are teaching values from the 2024 edition; local protocols and newer approvals take precedence.
- Speaker notes on some slides hold discussion prompts for the study group.

LEGEND FOR FLOWCHARTS



Solid box = step; dashed box = decision point; black box = preferred or default option. Arrows read left → right or top → bottom.

Parts and chapters

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Each Part opens with a divider slide; the appendix holds audit notes and abbreviations.

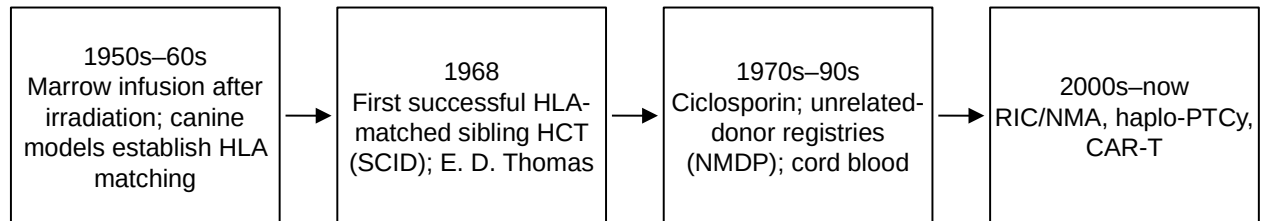
PART I · INTRODUCTION

Introduction

History of HCT, the EBMT organisation, donor registries, unit requirements, JACIE accreditation and transplant statistics.

Chapters 1–6

History of HCT



Why the milestones matter

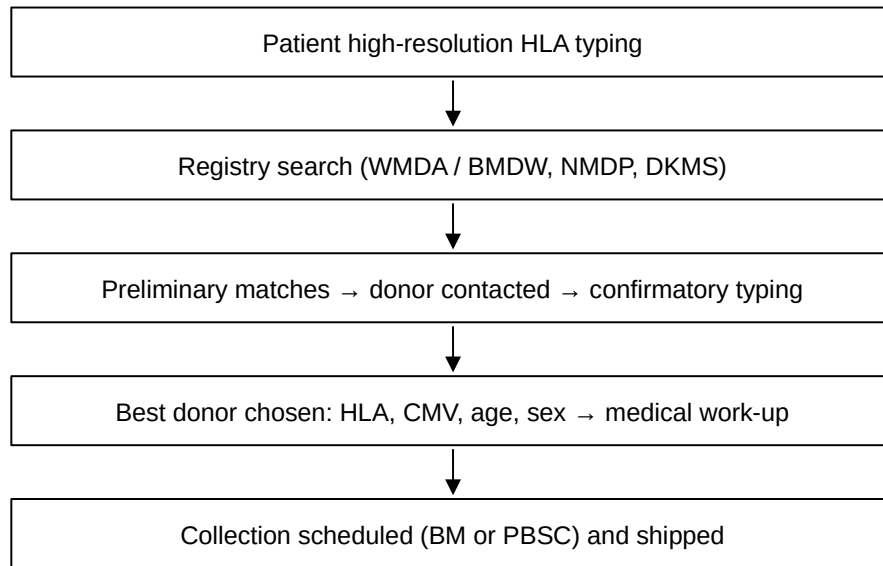
- Animal work (Storb, Thomas) showed histocompatibility, not just dose, determines engraftment and GVHD.
- Ciclosporin made GVHD controllable; registries and cord blood widened donor access.
- RIC/NMA conditioning extended eligibility beyond 70 years.
- Haploidentical PTCy made a donor available for almost every patient.
- CAR-T now competes with, and sometimes replaces, autologous HCT.

The EBMT: history, present and future

Aspect	Key facts
Founded	1974, Paris; started as an informal network of European transplant centres
Registry	Largest HCT registry worldwide; >45,000 transplants reported per year
Members	>600 centres in ~47 countries
Working Parties	ALWP, CLWP, PDWP, LWP, TCWP, IDWP and others produce guidelines and studies
Quality	JACIE accreditation (with ISCT) — see Ch. 5
Meeting	Annual Meeting each spring

Study point: the registry is the evidence base behind most EBMT recommendations quoted in this deck.

Donor registries: connecting donors and recipients



Scale

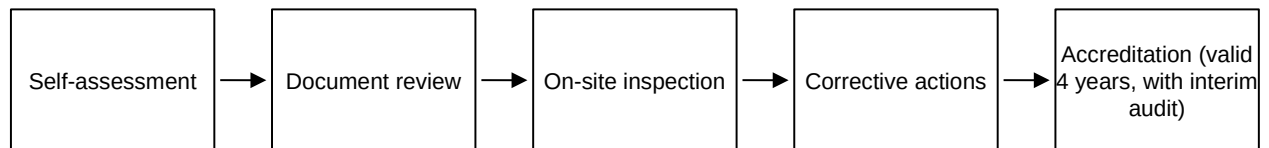
- > 40 million registered adult donors and ~800,000 cord blood units worldwide.
- DKMS is the largest single registry (> 12 million donors).
- Cord blood banks (Eurocord, NetCord) require minimum TNC and CD34+ content.

Requirements of an HCT unit

Domain	Requirement
Inpatient ward	HEPA-filtered, positive-pressure single rooms with own bathroom; visitor control
Blood bank	24-h supply; irradiated, leucodepleted / CMV-safe components
HLA laboratory	High-resolution typing of all relevant loci
Cell processing	GMP-grade collection and processing facility
Support	24-h radiology, ICU, microbiology, pharmacy
CAR-T add-ons	Apheresis, cold-chain logistics, CRS/ICANS expertise and equipment

JACIE accreditation

- **JACIE** = Joint Accreditation Committee ISCT–EBMT; European quality system, aligned with FACT.
- Covers three areas: **clinical programme, collection, processing**.
- Accredited centres show lower NRM and better survival in registry analyses.
- In several countries accreditation is a prerequisite for reimbursement or authorisation.



Statistical methods in HCT

Endpoint	Definition	Method
OS	Death from any cause	Kaplan–Meier; Cox
EFS / PFS / RFS	Relapse, progression or death	Kaplan–Meier; Cox
NRM	Death without relapse (competing with relapse)	Cumulative incidence; Fine–Gray
CIR	Relapse (competing with NRM)	Cumulative incidence; Gray's test
GRFS	Alive without relapse, grade III–IV aGVHD or severe cGVHD	Kaplan–Meier

Competing risks — the core idea

- Relapse and NRM compete: a patient who dies of GVHD can no longer relapse.
- 1 – Kaplan–Meier **overestimates** relapse because it censors competing deaths.
- Use cumulative incidence functions; landmark analysis for time-dependent covariates (e.g. GVHD).

PART II · BIOLOGICAL ASPECTS

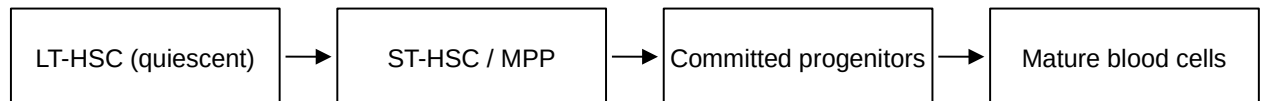
Biological Aspects

Stem cell and immune cell biology, histocompatibility and immune reconstitution after transplant.

Chapters 7–10

Biological properties of haematopoietic stem cells

- **Self-renewal** and multilineage differentiation define the HSC.
- **Long-term HSC (LT-HSC)**: quiescent, strongest self-renewal, maintained in the bone marrow niche.
- After HCT the donor HSC pool replaces host haematopoiesis (measured as chimerism, Ch. 21).
- LT-HSCs are the target for ex vivo gene therapy (e.g. exa-cel for SCD and thalassaemia).



Biological properties of non-HSC cells

Cell	Role after HCT	Clinical relevance
$\alpha\beta$ T cells	Main effectors of GVHD and GVL	Target of depletion strategies
$\gamma\delta$ T cells	Innate-like; retained after TCR $\alpha\beta$ /CD19 depletion	Lower infection risk without GVHD
Tregs	Suppress alloreactivity	Spared by PTCy → GVHD control
NK cells	Earliest lymphoid recovery (D+30–60); KIR-mediated alloreactivity	GVL in AML, esp. haplo
MSC	Immunomodulatory, tissue repair	Used off-label for steroid-refractory aGVHD

Histocompatibility

- **MHC class I** (HLA-A, -B, -C) presents intracellular peptides to CD8+ T cells; **class II** (HLA-DR, -DQ, -DP) presents extracellular peptides to CD4+ T cells.
- Matching for HCT: **HLA-A, -B, -C, -DRB1 (8/8) ± DQB1 (10/10)** at high resolution; DPB1 permissiveness increasingly considered.

Donor / match	GVHD risk	Impact on NRM / survival
10/10 matched sibling	Lowest	Reference
10/10 matched unrelated	Low	Slightly higher than MSD
9/10 mismatched unrelated	Intermediate	Each mismatch ≈ 10% lower survival
Haploidentical (5/10)	High without PTCy / T-cell depletion	Acceptable with PTCy platform

Non-HLA factors

- **KIR ligand mismatch:** NK alloreactivity, possible GVL benefit in AML.
- **Minor histocompatibility antigens:** why GVHD occurs even between HLA-identical siblings.
- **Female donor → male recipient** (H-Y antigens): more chronic GVHD.

Immune reconstitution

Compartment	Recovery	Clinical meaning
Neutrophils	D+14–28	First barrier; engraftment
NK cells	D+30–60	Early GVL
CD8+ T cells	3–6 months	CMV / EBV control
CD4+ T cells	6–12 months (slower after TBI)	Bacterial / fungal defence
B cells, immunoglobulins	12–24 months	Timing of re-vaccination

Slows reconstitution

- Ex vivo T-cell depletion
- Chronic GVHD and its treatment
- Older age (thymic involution)
- Cord blood graft

Monitoring (practical)

- Lymphocyte subsets at D+30, +100, +180
- IgG < 4 g/L with recurrent infection → IVIG replacement
- CD4 < 200/μL → continue PJP prophylaxis

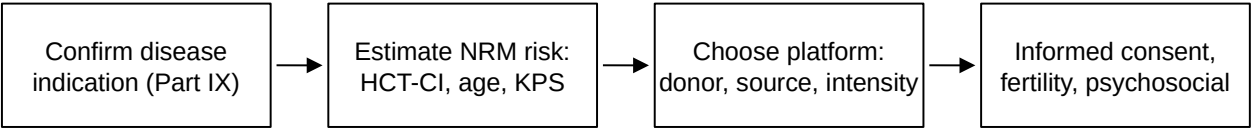
PART III · TRANSPLANT METHODOLOGY

Methodology and Clinical Aspects

Candidate evaluation, donor selection, conditioning, graft source, collection, cord blood, graft manipulation, processing, engraftment and chimerism.

Chapters 11–21

Evaluation and counselling of the candidate

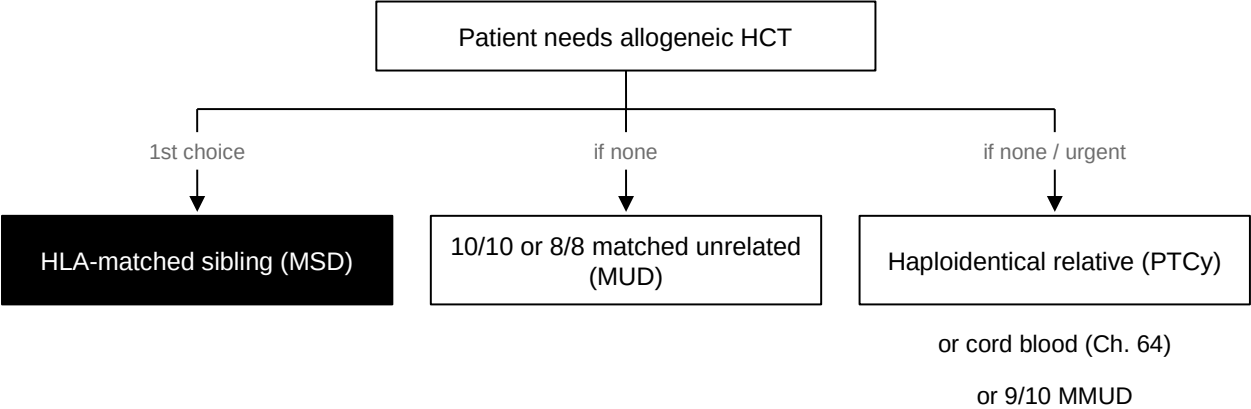


HCT-CI (Sorrow)	2-y NRM
0	≈ 14%
1–2	≈ 21%
≥ 3	≈ 41%

Organ	Typical minimum
Heart	LVEF ≥ 45%
Lung	FEV1 and DLCO ≥ 50%
Kidney	CrCl ≥ 40–50 mL/min
Liver	Bilirubin ≤ 2× ULN

- Other tools: **EBMT risk score** (age, stage, interval, donor, sex match), **KPS/ECOG**, geriatric assessment (Ch. 68).
- Infection screen: CMV, EBV, HSV, VZV, HBV, HCV, HIV, syphilis, TB (IGRA).

Donor selection



Secondary factor	Preference
Donor age	Younger (< 35–40 y) beats closer HLA match among unrelated donors
CMV serostatus	Match to recipient; R+/D– is the highest-risk constellation
Sex	Avoid female donor for male recipient (cGVHD)
ABO	Not decisive; guides transfusion plan (Ch. 24)
Donor-specific anti-HLA antibodies	Avoid in mismatched settings (graft failure)

Conditioning regimens

Intensity	Principle	Typical patient	Examples
MAC	Ablates host marrow; irreversible cytopenia	< 55–60 y, HCT-CI low	Bu/Cy, Cy/TBI 12 Gy, Flu/Bu4
RIC	Partial ablation; relies on GVL	55–70 y, comorbidity	Flu/Bu2, Flu/Mel 140, Flu/Treo
NMA	Minimal cytopenia; mixed chimerism	> 70 y or unfit	Flu/TBI 2 Gy, Flu/Cy

Reference doses (teaching values)

- **Bu/Cy:** busulfan 3.2 mg/kg/day IV × 4 + cyclophosphamide 60 mg/kg × 2.
- **Cy/TBI:** TBI 12 Gy fractionated (6 × 2 Gy) + cyclophosphamide 60 mg/kg × 2.
- **Flu/Bu RIC:** fludarabine 150 mg/m² + busulfan 6.4 mg/kg IV (8 mg/kg oral).
- **Flu/Mel:** fludarabine + melphalan 140 mg/m²; Seattle NMA = fludarabine 90 mg/m² + TBI 2 Gy.

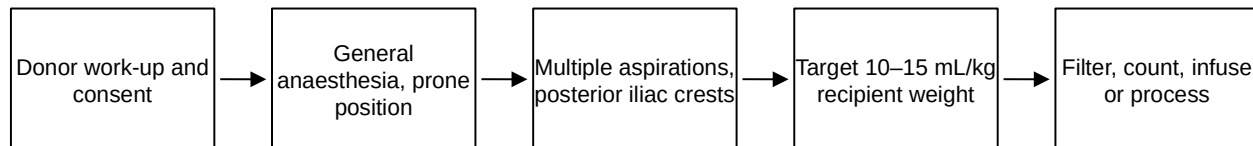
TBI late effects: pneumonitis, cataract, infertility, growth failure, secondary malignancy. Fractionation lowers toxicity.

Selecting the stem cell source

	Bone marrow	Peripheral blood	Cord blood
CD34+ dose	Intermediate	High	Low (but immature)
Neutrophil engraftment	Slower (~ +5 days)	Fastest	Slowest (D+25–28)
Chronic GVHD	Lower	Higher	Lowest
GVL	Standard	Stronger	Intermediate
Donor burden	GA, 100–200 punctures	G-CSF + apheresis	None
Preferred for	Non-malignant disease, children, aplastic anaemia	Adult malignancy	No suitable adult donor

- Adult malignancies: PBSC is now used in most transplants worldwide.
- Non-malignant or paediatric: BM preferred because GVL is not needed and cGVHD is pure harm.

Bone marrow harvesting



Targets

- $TNC \geq 2-3 \times 10^8/\text{kg}$ recipient
- $CD34+ \geq 2 \times 10^6/\text{kg}$ recipient
- Volume usually 500–1000 mL
- Do not exceed ~ 20 mL/kg donor weight

Donor safety

- Local pain in almost all donors, resolves in days
- Autologous back-up transfusion in ~ 10%
- Serious (mostly anaesthetic) events < 0.5%
- Indicated when BM is the preferred source (Ch. 14)

Mobilisation and collection of PBSC

Regimen	Schedule	Comment
G-CSF alone	10 µg/kg/day × 4–5 days; apheresis from day 5	Standard for healthy donors and many autos
G-CSF + plerixafor	Plerixafor 0.24 mg/kg s.c. evening of day 4 (pre-emptive or rescue)	For poor mobilisers; MM, lymphoma
Chemotherapy + G-CSF	e.g. cyclophosphamide 2–4 g/m ² then G-CSF	Highest yield, more toxicity

Target CD34+ dose

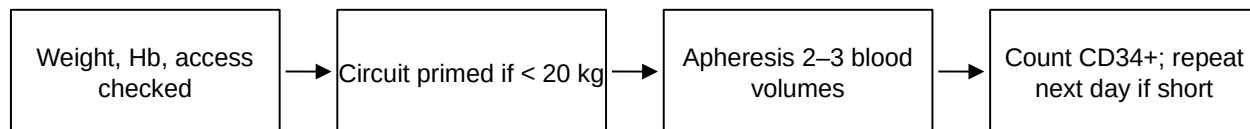
- Auto: ≥ 2 × 10⁶/kg minimum, ≥ 5 × 10⁶/kg optimal (more for tandem)
- Allo: ≥ 4 × 10⁶/kg recipient

Poor mobiliser (GITMO)

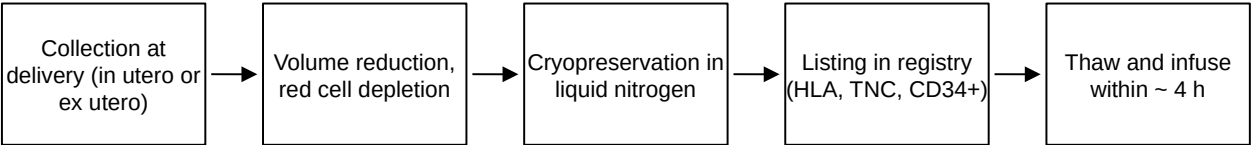
- Peak blood CD34+ < 20/µL up to day 6, or
- < 2 × 10⁶/kg in ≤ 3 aphereses
- Predictors: prior lenalidomide, fludarabine, extensive RT, marrow involvement

Mobilisation and collection in children

- Small blood volume: extracorporeal circuit is a large fraction of total volume → **prime the circuit with red cells** when < 20 kg (some centres < 25 kg).
- Vascular access usually needs a central apheresis catheter; watch for hypocalcaemia (citrate) and hypothermia.
- G-CSF 10 µg/kg/day for 4–5 days; start apheresis when blood CD34+ $\geq 10\text{--}20/\mu\text{L}$.
- Targets: $\geq 5 \times 10^6$ CD34+/kg for autologous (higher for tandem), $\geq 4 \times 10^6/\text{kg}$ for allogeneic.
- Below ~ 10 kg bone marrow harvest is often technically easier than apheresis.



Cord blood collection and banking



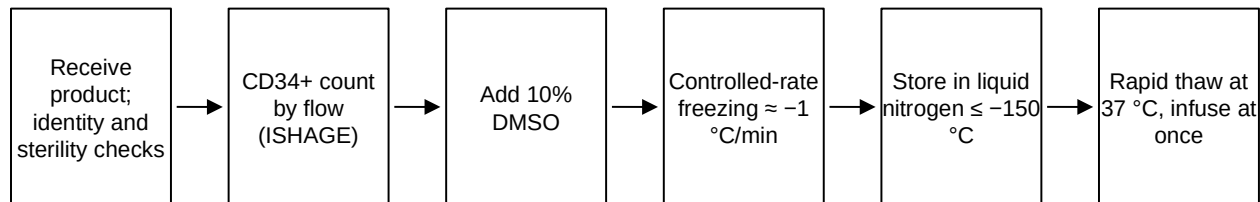
Unit selection	Threshold
HLA match	≥ 4/6 (A, B antigen; DRB1 allele); 8-allele high-resolution now preferred
TNC dose	≥ 3 × 10 ⁷ /kg recipient (single unit); cell dose outranks one extra mismatch
CD34+ dose	≥ 1.5 × 10 ⁵ /kg
Bank standards	Eurocord / NetCord–FACT; minimum ~ 80 mL collection, TNC ≥ 5 × 10 ⁸ per unit for banking

Graft manipulation: T-cell depletion

Platform	Principle	Residual T cells	Trade-off
CD34+ positive selection	Magnetic bead enrichment of stem cells	Very low (< 10 ⁴ /kg)	Slow immune recovery, infection, graft failure
TCRαβ / CD19 depletion	Removes αβ T and B cells; keeps γδ T, NK	Low	Preferred paediatric haplo platform
CD3 / CD19 depletion	Removes all T and B cells	Intermediate	Older approach
In vivo (ATG, alemtuzumab)	Serotherapy given to patient	Variable	Not graft manipulation but same goal

- Effect: GVHD ↓↓ but GVL ↓, infections ↑↑, graft failure ↑; often paired with add-back strategies.
- Regulatory: substantially manipulated grafts are ATMPs and need GMP facilities (Ch. 62).

Processing, cryopreservation and quality control



Release criteria

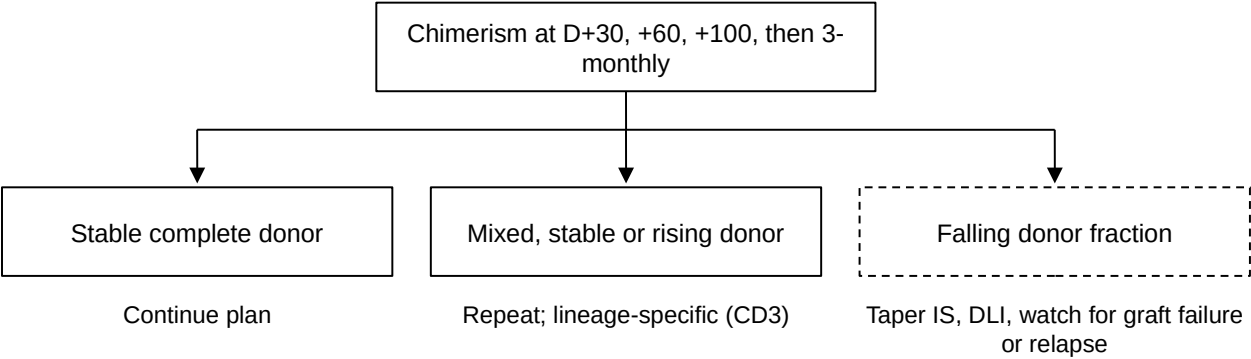
- Viability (7-AAD) after thaw $\geq 70\%$
- CFU-GM as functional assay
- Sterility, endotoxin, labelling
- Chain of identity documented

Infusion reactions

- DMSO: nausea, flushing, bradycardia, hypertension
- Dose-limit DMSO ≈ 1 g/kg/day
- Haemolysis from red cell debris (ABO)
- Pre-medicate; monitor vitals

Engraftment and chimerism monitoring

Term	Definition
Neutrophil engraftment	First of 3 consecutive days with ANC $\geq 0.5 \times 10^9/L$
Platelet engraftment	Platelets $\geq 20 \times 10^9/L$, 7 days transfusion-free
Complete chimerism	$\geq 95\%$ (or 100%) donor cells
Mixed chimerism	Donor $< 95\%$ with detectable recipient cells
Split chimerism	Different donor fractions in different lineages (e.g. T cells vs myeloid)



PART IV · GENERAL MANAGEMENT

General Management of the Patient

Follow-up, vascular access, transfusion, nutrition, GVHD prophylaxis, infection control, vaccination, psychology, drug interactions, nursing, ethics and quality of life.

Chapters 22–34

Short- and long-term follow-up after HCT

Time point	Main checks
D+30	Engraftment, chimerism, CMV/EBV PCR, CBC, CNI level
D+60	Chimerism, MRD (acute leukaemia), liver and renal function
D+100	Disease restaging, GVHD assessment, immune recovery, vaccine plan
D+180	Stop antiviral prophylaxis if eligible, CNI taper, start inactivated vaccines
1 year, then yearly	Organ late effects, secondary cancer screening, bone density, endocrine, QoL

Long-term organ surveillance

- Thyroid: TSH yearly after TBI
- Gonads: FSH/LH/oestradiol, testosterone, semen analysis (Ch. 56)
- Bone: DXA at 1 year, repeat if abnormal
- Eyes: cataract after TBI, sicca with cGVHD

Long-term general care

- Cardiovascular risk factors yearly (Ch. 55)
- Skin and oral cancer screening (Ch. 47)
- Revaccination schedule (Ch. 29)
- Psychosocial and return-to-work support (Ch. 30, 34)

Vascular access

Device	Features	Use in HCT
PICC (2–3 lumens)	Peripheral insertion, low insertion risk	Short term; thrombosis risk higher
Tunnelled CVC (Hickman/Broviac)	Cuffed, multi-lumen, long term	Standard for the transplant admission
Port-a-Cath	Fully implanted	Outpatient follow-up, intermittent use
Apheresis catheter	Large bore, high flow	Stem cell collection, ECP

Complications and prevention

- **CLABSI:** coagulase-negative staphylococci most common → vancomycin; remove catheter for S. aureus, Candida, persistent bacteraemia or tunnel infection.
- **Thrombosis:** LMWH once platelets allow; catheter may stay if functioning.
- **Mechanical:** pneumothorax, malposition, occlusion (alteplase lock).
- **Bundle:** ultrasound-guided insertion, chlorhexidine skin prep and dressings, daily review of need.

Transfusion support

Thresholds

- Red cells: Hb < 7–8 g/dL (restrictive)
- Platelets prophylactic: < 10 × 10⁹/L; < 20 if fever or bleeding risk
- Platelets with active bleeding or procedure: < 50 × 10⁹/L

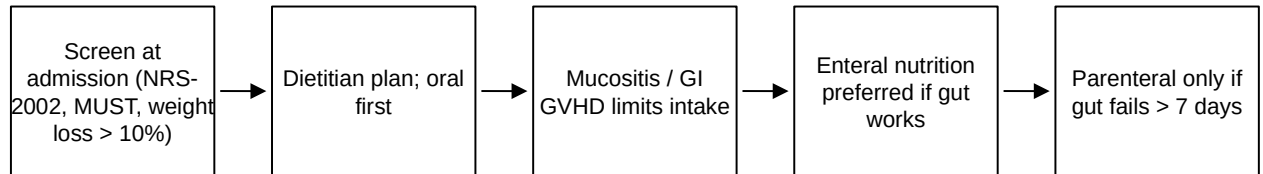
Product requirements

- **Irradiated** (≥ 25 Gy) cellular components for all HCT recipients: prevents TA-GVHD
- Leucodepleted or CMV-seronegative components for CMV-negative pairs
- Group-compatible plasma and platelets per ABO plan below

ABO mismatch	Meaning	Red cells	Platelets / plasma
Major	Recipient has antibody vs donor RBC (e.g. O → A)	Recipient type (O)	Donor type (A) or AB
Minor	Donor has antibody vs recipient RBC (e.g. A → O)	Donor type (O)	Recipient type (A) or AB
Bidirectional	Both (e.g. A → B)	Group O	Group AB

Major mismatch: risk of pure red cell aplasia and delayed red cell engraftment; consider red cell depletion of a marrow graft. Minor mismatch: passenger lymphocyte haemolysis around D+7–14.

Nutritional support



- Energy 25–30 kcal/kg/day; protein 1.5–2.0 g/kg/day in the acute phase.
- Enteral feeding is associated with less infection and better outcomes than routine TPN.
- Low-microbial ('neutropenic') diets lack strong evidence; focus on safe food handling and avoiding high-risk raw items.
- Monitor glucose, triglycerides and electrolytes on TPN; refeeding risk in malnourished patients.

GVHD prophylaxis

Setting	Standard backbone	Common additions
MAC, MSD or MUD	Calcineurin inhibitor (CSA or tacrolimus) + short-course MTX	ATG for unrelated donors (cGVHD ↓)
RIC, MUD	CNI + MMF	ATG
Haploidentical	PTCy 50 mg/kg D+3 and D+4, then tacrolimus + MMF from D+5	—
Emerging / trial-based	PTCy also in matched donors (BMT CTN 1703)	Abatacept, vedolizumab (GI protection), sirolimus

ATG principles

- Rabbit ATG (Thymoglobulin) ≈ 4.5–6 mg/kg total; ATG-Fresenius (Grafalon) 30–60 mg/kg — products are not interchangeable.
- Reduces acute and chronic GVHD without clear survival gain; delays immune recovery, more viral reactivation.
- Target the dose to lymphocyte count and weight in children.

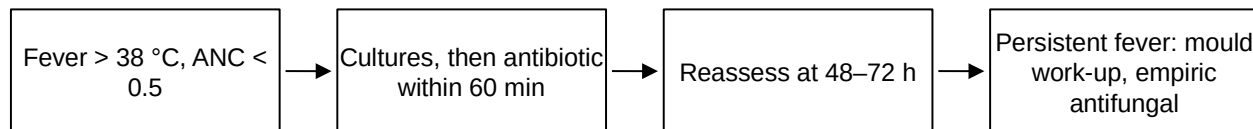
MTX: 15 mg/m² D+1 then 10 mg/m² D+3, +6 (±11). Leucovorin rescue and dose reduction for mucositis or renal impairment.

Infection control and isolation

Measure	Detail
Hand hygiene	Alcohol rub before and after every contact — the single most effective measure
Environment	HEPA filtration, positive pressure, ≥ 12 air changes/h; no plants or flowers; avoid construction dust (Aspergillus)
Precautions	Contact isolation for MDRO carriers; droplet for respiratory viruses; airborne (N95) for TB, measles, varicella
Screening	Admission rectal swab (ESBL, CPE, VRE) and nasal MRSA per local policy
Visitors	Limited, screened for symptoms; no ill visitors
Food	Safe handling; avoid raw meat, fish, unpasteurised dairy, unwashed produce

Supportive care of infection in children

- Same principles as adults; **dose by weight or BSA** and expect different pharmacokinetics.
- Febrile neutropenia first line: piperacillin-tazobactam (≈ 300 mg/kg/day of piperacillin, divided q6–8h) or cefepime.
- Antifungal prophylaxis: fluconazole in young children; mould-active azoles (voriconazole, posaconazole) in high-risk adolescents — voriconazole PK is highly variable in children → frequent TDM.
- Liposomal amphotericin B is well tolerated in children and useful when azoles fail or interact.
- Viruses: CMV, EBV and especially **adenovirus** cause more severe disease after haplo or T-cell-depleted paediatric HCT; monitor by PCR, treat with cidofovir (brincidofovir where available).



Vaccination after HCT

Vaccine	Start	Notes
Pneumococcal conjugate (PCV)	3–6 months	3 doses, then PPSV23 or 4th PCV at 12 months (PCV if cGVHD)
Inactivated influenza	6 months (4 in outbreak)	Yearly, lifelong; household contacts too
COVID-19 (mRNA)	3 months	Full primary series again
DTaP, Hib, IPV, HBV	6–12 months	3 doses each
HPV	6–12 months	3 doses, all sexes
MMR (live)	24 months	Only if no active cGVHD, off IS, no IVIG in 8–11 months
Varicella (live)	24 months	Same conditions; recombinant zoster vaccine can be used earlier

Live vaccines are contraindicated until immune reconstitution (usually ≥ 24 months, no GVHD, no immunosuppression). Household members: yearly influenza and age-appropriate live vaccines are safe.

Psychological aspects

Phase	Typical problems	Approach
Pre-transplant	Anxiety, uncertainty, fear of procedures	Screen (HADS, distress thermometer), psycho-education, CBT
Inpatient	Isolation, body-image change, loss of control, delirium	Daily structure, family contact within infection rules, early mobilisation
Post-transplant	Fatigue (most persistent), depression, PTSD, sexual problems	Rehabilitation, graded activity, psychotherapy, couple counselling
cGVHD	Highest symptom and psychological burden	Integrated multidisciplinary clinic
Adolescents / young adults	Interrupted education, peers, identity, fertility	AYA-specific programmes, fertility counselling before conditioning

Clinically relevant drug interactions

Raise CNI / sirolimus level (CYP3A4 inhibitors)	Lower CNI level (CYP3A4 inducers)
Voriconazole, posaconazole, itraconazole, isavuconazole (weaker)	Rifampicin, rifabutin
Fluconazole (moderate, dose-dependent)	Carbamazepine, phenytoin, phenobarbital
Clarithromycin, erythromycin	St John's wort
Letermovir (raises tacrolimus and CSA; CSA raises letermovir)	Nafcillin, dexamethasone (mild)
Grapefruit, diltiazem, verapamil	—

Practical rules

- Starting voriconazole or posaconazole: cut tacrolimus to $\approx 1/3$ (CSA to $\approx 1/2$) and measure levels within 3–5 days; reverse on stopping.
- Rifampicin can make tacrolimus levels undetectable — avoid or use rifabutin with close TDM.
- QT prolongation: azoles + ondansetron, fluoroquinolones, methadone → ECG and electrolytes.
- Busulfan with metronidazole or itraconazole raises Bu exposure; separate them.
- MMF levels fall with CSA (enterohepatic cycle) but not with tacrolimus.

The role of nursing in HCT

Role	Core responsibilities
Transplant coordinator	Donor search liaison, work-up scheduling, timeline, single point of contact for patient and family
Apheresis nurse	Runs collection, manages citrate toxicity and access problems, product labelling
Ward nurse	Mucositis and skin care, CVC dressing under aseptic technique, transfusion monitoring, first to spot GVHD rash or diarrhoea
Advanced practice nurse	Leads follow-up clinics, symptom management, survivorship, protocol adherence
Education role	Teaches self-care: temperature, hygiene, medication, when to call

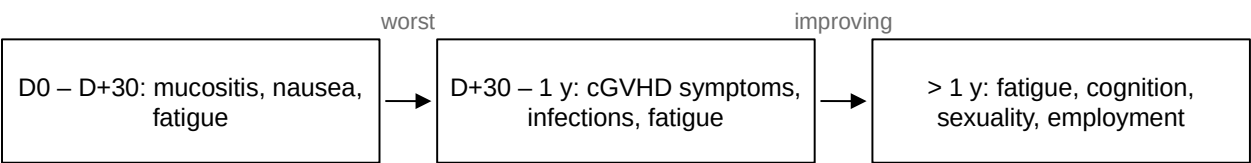
JACIE standards require documented nurse training and competency for HCT and cellular therapy programmes.

Ethical issues in HCT

- **Informed consent** for a high-risk therapy: NRM, GVHD, infection, infertility, secondary cancer must be discussed; urgency must not shortcut understanding.
- **Donor protection**: voluntary, informed, free to withdraw; donor advocate independent of the recipient's team; minor sibling donors need ethics review and assent.
- **Cord blood**: public banking (altruistic) vs private banking (limited evidence of benefit).
- **Resource allocation**: cost of HCT and ATMPs vs equity; access to clinical trials.
- **End of life**: when to stop escalating (second transplant, ICU); advance care planning belongs in the pre-transplant consultation.

Quality of life assessment

Instrument	Population	Focus
FACT-BMT	Adult HCT	HCT-specific module of FACT-G
EORTC QLQ-C30 (+ HDC29)	Adult oncology	Generic cancer QoL, HCT add-on module
PedsQL	Children	Child- and parent-reported
Lee cGVHD Symptom Scale	cGVHD	30 symptoms across 7 domains
PROMIS / PRO-CTCAE	Any	Patient-reported symptoms in trials and clinics



PART V · COMPLICATIONS

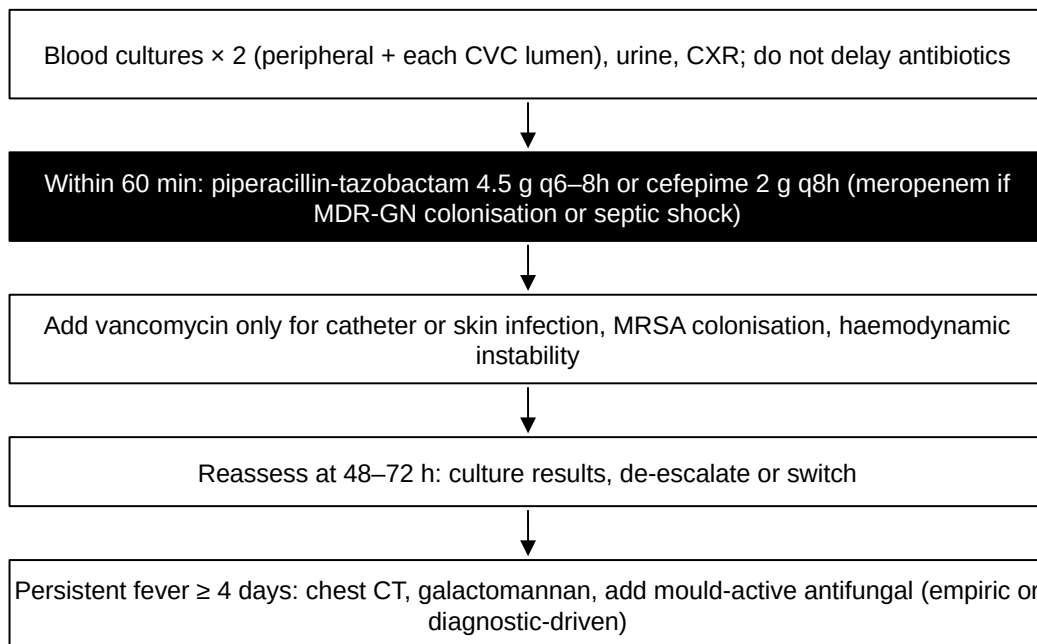
HCT Complications

Febrile neutropenia, bacterial, fungal, viral and other infections, bleeding and thrombosis, graft failure, endothelial syndromes, acute and chronic GVHD, PTLN, iron overload and secondary malignancies.

Chapters 35–47

Febrile neutropenia

- Definition: temperature ≥ 38.3 °C once or ≥ 38.0 °C for 1 h with ANC $< 0.5 \times 10^9/L$. **Every HCT recipient counts as high risk.**



Stop empirical antibiotics after 72 h if afebrile ≥ 48 h, cultures negative and patient stable, even before neutrophil recovery (ECIL-4). Check the calculator: `scripts/hct_tools.py print_checklist('fn')`.

Bacterial infections

Organism	Typical setting	Treatment
Coagulase-negative staphylococci	CVC, most frequent bacteraemia	Vancomycin; often keep catheter
S. aureus / MRSA	CVC, skin	Vancomycin (daptomycin, ceftaroline alt.); remove CVC
VRE	Prolonged hospitalisation, prior vancomycin	Linezolid or daptomycin
Pseudomonas	Mucositis, neutropenia	Pip-tazo, cefepime or meropenem; ceftolozane-tazobactam if resistant
ESBL Enterobacterales	Colonisation	Carbapenem
KPC carbapenemase	Colonisation, outbreaks	Ceftazidime-avibactam or meropenem-vaborbactam
Metallo-β-lactamase (NDM, VIM)	Colonisation	Ceftazidime-avibactam + aztreonam, or cefiderocol
C. difficile	Antibiotics, PPI	Oral vancomycin or fidaxomicin

- Bacteraemia: treat 7–14 days from first negative culture; longer for S. aureus.
- Fluoroquinolone prophylaxis during neutropenia is used in many centres but drives resistance — decide locally.

Invasive fungal infections

Risk

- High: allo-HCT, neutropenia > 10 days, high-dose steroids, GVHD, cord blood, CMV
- Intermediate: auto-HCT for lymphoma or myeloma
- Agents: *Aspergillus fumigatus* (most), *Candida*, *Mucorales*, *Fusarium*

Diagnosis

- Chest CT: halo sign early, air-crescent late
- Galactomannan: BAL more sensitive than serum; falsely low on mould-active prophylaxis
- β -D-glucan: broad, negative in mucormycosis
- *Aspergillus* PCR, culture and histology when possible

Situation	Drug	Dose
Prophylaxis, low mould risk	Fluconazole	400 mg/day (no mould cover)
Prophylaxis, GVHD on steroids	Posaconazole	300 mg/day (after loading)
Invasive aspergillosis 1st line	Voriconazole	6 mg/kg q12h $\times$ 2, then 4 mg/kg q12h; TDM 1–5.5 mg/L
Alternatives	Isavuconazole or liposomal amphotericin B	L-AmB 3 mg/kg/day
Mucormycosis	L-AmB + surgical debridement	5–10 mg/kg/day; isavuconazole or posaconazole step-down

Viral infections

Virus	Monitoring	Prevention / treatment
CMV	Weekly PCR to D+100 (D+180–200 if high risk)	Letermovir prophylaxis in R+ to D+100 (D+200 high risk). Pre-emptive: ganciclovir 5 mg/kg q12h or foscarnet. Refractory/resistant: maribavir
EBV	Weekly PCR if ATG, T-cell depletion, haplo, cord	Rising DNA → rituximab 375 mg/m ² pre-emptive; PTLD see Ch. 45
HHV-6	PCR when encephalitis suspected	Foscarnet or ganciclovir; limbic encephalitis around engraftment
Adenovirus	PCR in children, TCD, haplo	Reduce IS; cidofovir; virus-specific T cells
BK virus	Urine/plasma PCR if haematuria	Haemorrhagic cystitis (Ch. 51)
RSV, influenza, SARS-CoV-2	Respiratory PCR	Treat upper-tract disease early: ribavirin ± IVIG (RSV), oseltamivir, nirmatrelvir/remdesivir
HSV / VZV	—	Aciclovir prophylaxis for all seropositives ≥ 1 year (longer with cGVHD)

Highest CMV risk: seropositive recipient with seronegative donor (R+/D–), haplo-PTCy, cord blood, ATG, steroids.

Other severe infections: toxoplasma, TB, NTM

Infection	Who is at risk	Screen / prevent	Treat
Toxoplasmosis	Seropositive recipient, cord, TCD, steroids	Serology pre-HCT; TMP-SMX (also covers PJP); PCR monitoring if high risk	Pyrimethamine + sulfadiazine + folinic acid; TMP-SMX alternative
Tuberculosis	Endemic exposure, prior TB	IGRA (preferred over TST); latent TB → isoniazid 6–9 months	Standard 4-drug therapy; rifampicin crashes CNI levels (Ch. 31)
Non-tuberculous mycobacteria	CVC, cGVHD, lung disease	None routine	Species-directed (e.g. macrolide + ethambutol + rifabutin for MAC); remove catheter
Pneumocystis	All allo, auto with steroids	TMP-SMX from engraftment ≥ 6 months (longer on IS)	High-dose TMP-SMX ± steroids
Strongyloides	Tropical exposure	Serology; ivermectin before conditioning	Ivermectin

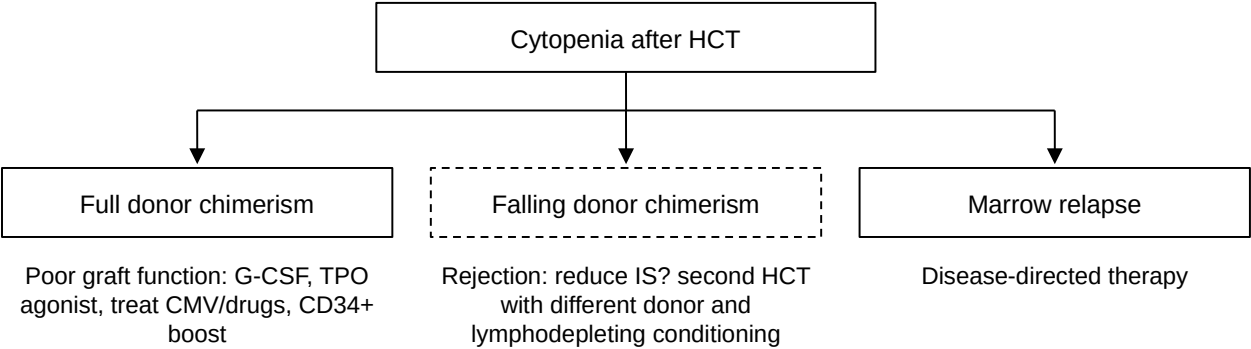
Bleeding and thrombotic complications

Problem	Risk factors	Management
Catheter-related DVT / PE	CVC, immobility, cancer, prior VTE	LMWH; hold if platelets < 50 (< 30 with caution); keep working catheter
TA-TMA	CNI, sirolimus, CMV, GVHD, complement variants	Stop CNI, treat trigger, eculizumab (Ch. 42)
SOS/VOD hepatic outflow	See Ch. 49	Defibrotide
Bleeding	Thrombocytopenia, mucositis, GVHD, coagulopathy	Platelets to > 50 if bleeding, fibrinogen > 1.5 g/L, treat cause
Diffuse alveolar haemorrhage	Engraftment, conditioning	High-dose steroids (Ch. 52)
Haemorrhagic cystitis	Cyclophosphamide, BK virus	Hydration, irrigation (Ch. 51)

- Thrombocytopenia is the dominant cause of bleeding; prophylactic platelets at < 10 × 10⁹/L (Ch. 24).
- Pharmacological VTE prophylaxis is generally withheld while platelets are < 50 × 10⁹/L; mechanical prophylaxis instead.

Graft failure

Term	Definition
Primary graft failure	No neutrophil engraftment (ANC < 0.5 × 10 ⁹ /L) by D+28 (BM/PBSC) or D+42 (cord blood)
Secondary graft failure	Loss of donor haematopoiesis after initial engraftment
Poor graft function	Cytopenias with full donor chimerism (no rejection)
Graft rejection	Loss of donor cells with recovery of recipient haematopoiesis (immune-mediated)



Risk factors: HLA mismatch, donor-specific antibodies, low cell dose, T-cell depletion, RIC in non-malignant disease, viral infection (CMV, HHV-6), myelotoxic drugs.

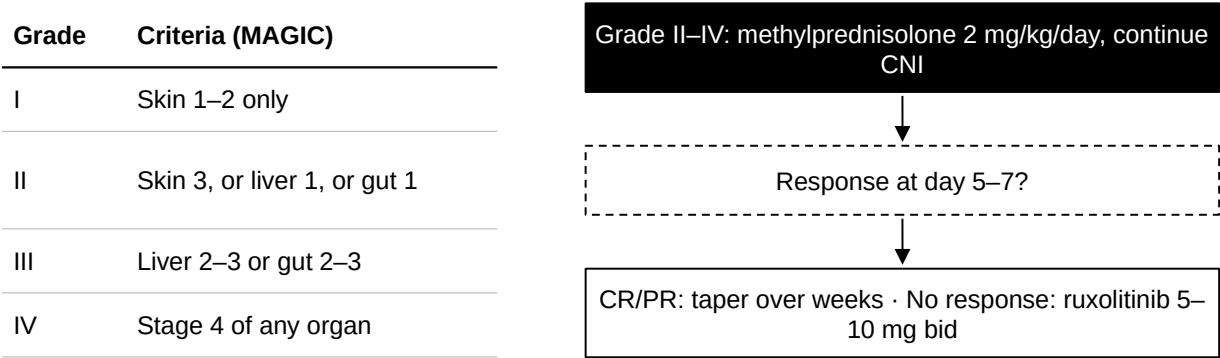
Early complications of endothelial origin

Syndrome	Timing	Key features	Treatment
SOS / VOD	< 21 d (late forms exist)	Bilirubin ↑, painful hepatomegaly, weight gain, ascites	Defibrotide (Ch. 49)
Engraftment syndrome / capillary leak	Around ANC recovery	Fever, rash, pulmonary infiltrates, weight gain	Steroids, diuretics
TA-TMA	Weeks to months	Schistocytes, LDH ↑, thrombocytopenia, proteinuria, hypertension, renal injury	Stop CNI/sirolimus, treat trigger, eculizumab
Diffuse alveolar haemorrhage	D+10 – D+30	Hypoxia, bloody BAL without infection	High-dose steroids
PRES	Any, CNI-related	Headache, seizures, visual loss, posterior oedema on MRI	Switch/stop CNI, control BP

- Common mechanism: endothelial activation by conditioning, cytokines, CNI, infection and alloreactivity.
- Endothelial Activation and Stress Index (EASIX = $\text{LDH} \times \text{creatinine} / \text{platelets}$) predicts these syndromes.

Acute GVHD

Organ	Stage 1	Stage 2	Stage 3	Stage 4
Skin (rash)	< 25% BSA	25–50% BSA	> 50% BSA	Generalised erythroderma with bullae / desquamation
Liver (bilirubin)	2–3 mg/dL	3.1–6 mg/dL	6.1–15 mg/dL	> 15 mg/dL
Gut (stool volume)	500–999 mL/day or persistent nausea	1000–1500 mL/day	> 1500 mL/day	Severe pain, ileus, or grossly bloody stool



Classic aGVHD ≤ D+100; late-onset aGVHD > D+100 (after RIC or DLI). Steroid-refractory: ruxolitinib (REACH2) is first choice; alternatives ECP, MMF, etanercept, ATG, MSC, faecal transplant in trials. Calculator: `hct_tools.grade_acute_gvhd()`.

Chronic GVHD

Epidemiology and risk

- 30–70% of allo-HCT; leading cause of late NRM
- Prior aGVHD, PBSC graft, unrelated or mismatched donor, older age, female → male, no ATG/PTCy

NIH 2014 diagnosis

- ≥ 1 diagnostic sign (poikiloderma, lichen planus-like, sclerosis, oesophageal web, BOS) or
- ≥ 1 distinctive sign plus biopsy or test confirmation
- Organs: skin, mouth, eyes, GI, liver, lung, joints/fascia, genital

NIH global severity	Rule
Mild	1–2 organs, each score 1 (lung 0)
Moderate	≥ 3 organs score 1, or any organ score 2, or lung score 1
Severe	Any organ score 3, or lung score ≥ 2

Line	Options
First	Prednisone 1 mg/kg/day ± CNi; taper over 6–12 months; topical therapy for limited disease
Second	Ruxolitinib 10 mg bid (REACH3), belumosudil 200 mg/day, ibrutinib 420 mg/day, ECP (skin, mouth, liver), MMF, sirolimus, imatinib for sclerosis
Supportive	Infection prophylaxis, IVIG if hypogammaglobulinaemia, physiotherapy, eye and dental care, bone protection

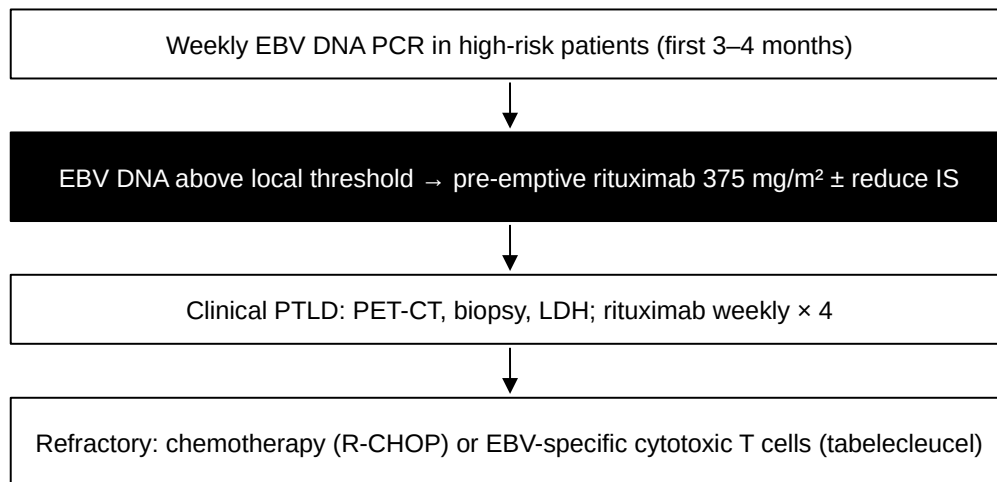
Post-transplant lymphoproliferative disorders

Classification (WHO)

- Non-destructive lesions (plasmacytic hyperplasia, IM-like)
- Polymorphic PTLD
- Monomorphic PTLD — DLBCL most common
- Classic Hodgkin-type PTLD

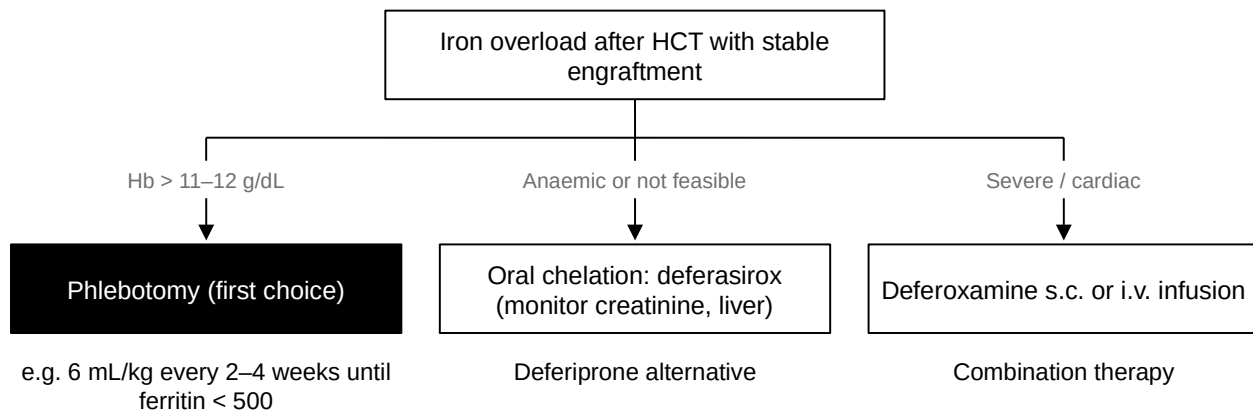
Risk factors

- T-cell depletion, ATG, alemtuzumab
- HLA mismatch, haplo, cord blood
- EBV-seronegative recipient with seropositive donor
- Severe GVHD and intense immunosuppression



Iron overload

- Source: transfusion ($\approx 200\text{--}250$ mg iron per red cell unit) plus ineffective erythropoiesis; no physiological excretion.
- Pre-HCT iron overload associates with more infection (*Aspergillus*, mucormycosis), SOS/VOD and NRM — ferritin > 1000 ng/mL is a crude marker (inflammation confounds).
- Quantify with **MRI (T2* or R2) liver iron concentration**; transferrin saturation; cardiac MRI if very high burden.



Secondary malignancies

Type	Main risk factors	Timing	Screening
PTLD (EBV)	T-cell depletion, mismatch	First year	EBV PCR (Ch. 45)
Therapy-related MDS/AML	Alkylators, etoposide, TBI; mostly after auto-HCT	2–10 years	CBC; marrow if cytopenias
Solid tumours: skin (SCC, BCC), oral, thyroid, breast, oesophagus, liver	TBI, cGVHD, prolonged IS, HPV, young age at TBI	> 5–10 years, rising with time	Yearly skin and oral exam, TSH, mammography from age 25 or 8 y after chest RT, HPV vaccination, avoid sun and tobacco
Donor cell leukaemia	Rare	Any	Chimerism plus morphology

- Cumulative incidence of solid cancers $\approx$ 2–3× the general population, still rising at 20 years.
- Survivorship visits should include cancer screening at least as intensive as age-matched population guidelines.

PART VI · ORGAN-SPECIFIC COMPLICATIONS

Organ-Specific Complications

Eye and mouth, liver and SOS/VOD, gut, bladder and kidney, lung, nervous system, skin and musculoskeletal, cardiovascular and metabolic, endocrine and fertility.

Chapters 48–56

Ocular and oral complications

Eye problem	Clues	Management
Dry eye (ocular cGVHD)	Burning, photophobia; Schirmer ≤ 5 mm/5 min, OSDI > 23	Preservative-free tears, punctal plugs, topical ciclosporin, scleral lenses, ECP
Cataract	After TBI or steroids	Ophthalmology; surgery
CMV retinitis	Floaters, field loss	Intravenous ganciclovir $\pm$ intravitreal
Orbital / sinus mould	Proptosis, pain	L-AmB + surgery

Oral problem	Phase	Management
Mucositis	Peak D+7–14	Oral care protocol, cryotherapy (melphalan), palifermin (auto-HCT with TBI), opioid PCA
HSV reactivation	Early	Aciclovir prophylaxis for all seropositive patients
Oral cGVHD	Late	Topical clobetasol or tacrolimus, dexamethasone rinse; pilocarpine for xerostomia; yearly cancer screen
Dental	Pre-HCT and yearly	Clear infection foci before conditioning

Hepatic complications and SOS/VOD

EBMT adult criteria (2016)

- Classical (≤ 21 days): bilirubin ≥ 2 mg/dL **plus two of** painful hepatomegaly, weight gain $> 5\%$, ascites
- Late-onset (> 21 days): classical criteria, or histology, or ≥ 2 criteria with ultrasound/haemodynamic evidence
- 2023 refined criteria add imaging and remove the strict day-21 cut-off

Risk factors

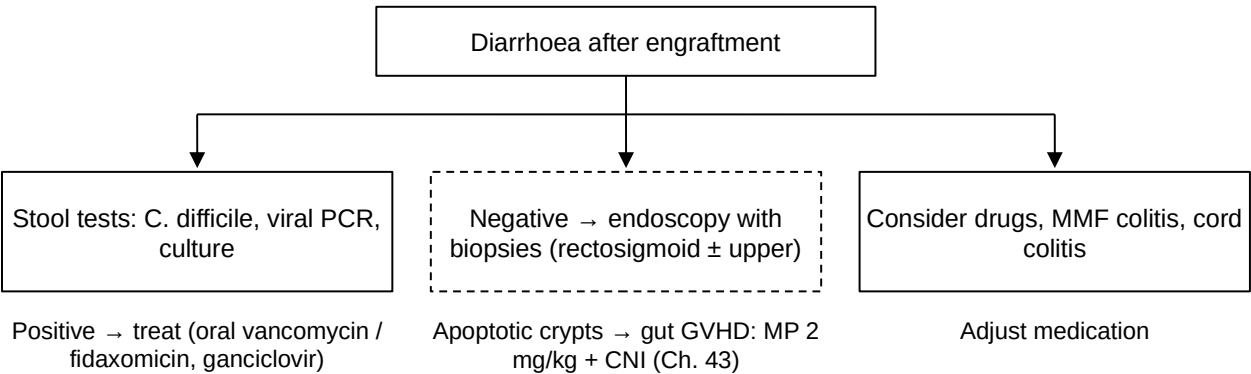
- Busulfan-based MAC, TBI
- Gemtuzumab or inotuzumab before HCT
- Second HCT, pre-existing liver disease, iron overload
- SCD, thalassaemia, osteopetrosis, HLH in children (incidence 20–40%)
- Sirolimus, norethisterone

Severity	Bilirubin kinetics / organ failure	Action
Mild	Bilirubin ≥ 2 and < 3 , slow rise, no organ failure	Supportive
Moderate	3–5 mg/dL or rise over 2–5 days	Consider defibrotide
Severe	5–8 mg/dL, doubling within 48 h, renal or lung involvement	Defibrotide now
Very severe	> 8 mg/dL or multi-organ failure	Defibrotide; ICU; mortality $> 80\%$

- **Defibrotide** 6.25 mg/kg q6h (25 mg/kg/day) for ≥ 21 days — only approved drug; fluid restriction, diuretics, avoid nephrotoxins; no benefit shown for prophylaxis in adults (HARMONY).
- Differentials: drug hepatotoxicity, hepatic GVHD, viral hepatitis, cholestasis of sepsis, TPN.

Gastrointestinal complications

Timing	Main causes
Conditioning	Mucositis, chemotherapy/TBI-induced nausea and vomiting
D0 – D+30	Infectious enteritis (C. difficile, CMV, norovirus, adenovirus), acute GVHD, drug toxicity (MMF, MTX)
> D+30	Chronic GVHD (oesophageal web, strictures, malabsorption), infections, pancreatic insufficiency



- Antiemetics: 5-HT3 antagonist + dexamethasone + NK1 antagonist for highly emetogenic conditioning; olanzapine as add-on.
- GI bleeding: thrombocytopenia, GVHD ulcers, CMV; support with transfusion and treat cause.

Haemorrhagic cystitis and renal dysfunction

Haemorrhagic cystitis

- **Early** (D0–7): cyclophosphamide/ifosfamide acrolein → prevent with mesna and hyperhydration
- **Late** (≥ D+14): BK virus (most), adenovirus, CMV; aggravated by GVHD and immune recovery
- Grade 1 microscopic; 2 macroscopic; 3 clots; 4 obstruction / renal failure

Management

- Hyperhydration, urine output > 100 mL/h, platelets > 50, analgesia
- Continuous bladder irrigation for clots (grade 3–4)
- Reduce IS; cidofovir (nephrotoxic; intravesical option), BK-specific T cells in trials
- Refractory: hyperbaric oxygen, intravesical agents, embolisation, cystectomy

Renal problem	Causes	Notes
Early AKI (up to 50%)	Sepsis, CNI, amphotericin, aminoglycosides, SOS/VOD hepatorenal, TMA	AKI needing dialysis: mortality > 50%
Chronic kidney disease	CNI, TA-TMA, TBI nephropathy, BK nephropathy	Control BP, avoid nephrotoxins, monitor eGFR yearly
Nephrotic syndrome	Membranous GN as renal cGVHD	Steroids + CNI/rituximab

Non-infectious pulmonary complications

Early (< D+100)	Timing	Features	Treatment
Idiopathic pneumonia syndrome (IPS)	D+7 – D+60	Diffuse infiltrates, hypoxia, no pathogen on BAL	High-dose steroids ± etanercept
Diffuse alveolar haemorrhage	D+10 – D+30	Progressively bloodier BAL returns	High-dose steroids
Engraftment syndrome / PERDS	Neutrophil recovery	Fever, rash, oedema, infiltrates	Steroids
Late (> D+100)	Features		Treatment
Bronchiolitis obliterans syndrome (BOS)	Only diagnostic lung cGVHD sign: FEV1/FVC < 0.7, FEV1 < 75%, air trapping (RV > 120% or expiratory CT), no infection		FAM (fluticasone, azithromycin, montelukast), ruxolitinib, belumosudil, ECP; lung transplant
Cryptogenic organising pneumonia	Patchy consolidation, steroid-responsive		Prednisone taper over months
Pleuroparenchymal fibroelastosis	Upper-lobe fibrosis, pneumothorax		Supportive

Screen with spirometry every 3 months in the first 2 years and at any respiratory symptom; a 10% FEV1 drop warrants CT and BAL.

Neurological complications

Category	Examples	Clues
Drug toxicity	CNI → PRES, tremor; busulfan seizures; MTX / ifosfamide encephalopathy; fludarabine leukoencephalopathy	Timing, levels, MRI pattern
Infection	HHV-6 limbic encephalitis (most common), Aspergillus, Toxoplasma, Listeria, JC virus (PML), VZV vasculitis	PCR in CSF, imaging
Vascular	Haemorrhage (thrombocytopenia), TMA-related encephalopathy	Sudden onset
Immune	CNS GVHD (rare), myasthenia, CIDP	Exclusion diagnosis
Metabolic	Hyponatraemia, hypomagnesaemia, hepatic encephalopathy	Labs

PRES

- Headache, seizures, visual disturbance, hypertension
- MRI: posterior vasogenic oedema (T2/FLAIR)
- Stop or switch CNI, treat BP; usually reversible

HHV-6 encephalitis

- Around engraftment (2–6 weeks), cord blood risk
- Amnesia, confusion, seizures; hippocampal T2 signal
- Foscarnet 90 mg/kg q12h or ganciclovir; sequelae common

Skin, hair and musculoskeletal complications

Condition	Features	Management
Skin aGVHD	Maculopapular rash starting palms, soles, ears, nape; stage 4 = bullae	Topical steroids for stage 1–2; systemic per Ch. 43
Lichenoid cGVHD	Violaceous flat papules, poikiloderma	Topical clobetasol/tacrolimus, phototherapy, systemic therapy
Sclerotic cGVHD	Skin thickening, fasciitis, joint contractures	Early physiotherapy, ECP, ruxolitinib, imatinib, belumosudil
Avascular necrosis	Femoral head; steroids, older age; MRI early (X-ray negative)	Load reduction, analgesia, joint replacement
Osteoporosis	Steroids, hypogonadism, CNI	Ca + vitamin D, bisphosphonate, DXA at 1 year
Steroid myopathy	Proximal weakness	Taper steroids, physiotherapy

- Hair: anagen effluvium after conditioning regrows in 3–6 months; permanent alopecia after busulfan or with scalp cGVHD.
- Nails and hair changes are distinctive cGVHD signs; photograph and score joints (P-ROM) at each visit.

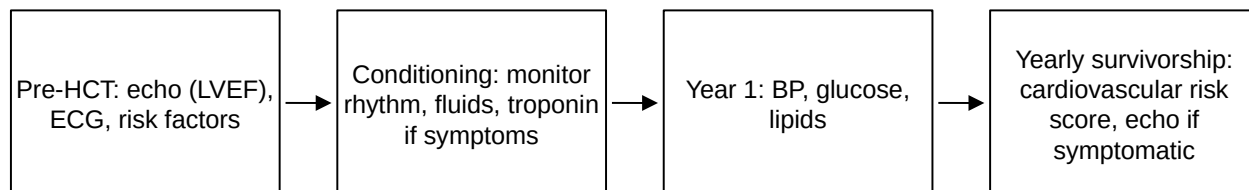
Cardiovascular and metabolic complications

Cardiac

- Cyclophosphamide cardiotoxicity (high dose, early, haemorrhagic myocarditis)
- Prior anthracycline: cumulative dose drives late heart failure
- Arrhythmias during conditioning and sepsis; QT prolongation (azoles, ondansetron)
- Hypertension from CNI and steroids
- Long-term: 2–4× cardiovascular events vs population

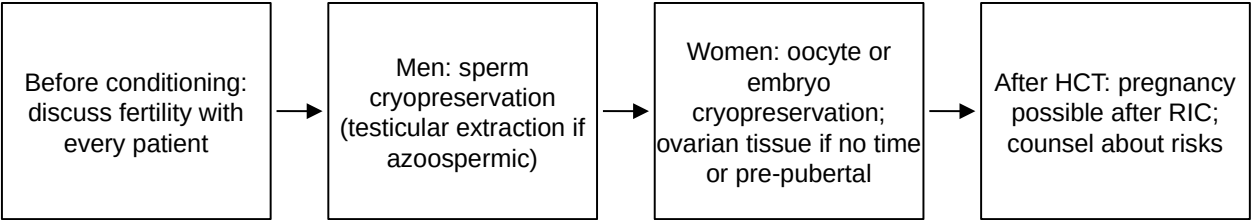
Metabolic syndrome

- Central obesity, dysglycaemia, dyslipidaemia, hypertension
- Drivers: steroids, CNI, TBI, hypogonadism, inactivity
- Screen yearly: BP, fasting glucose or HbA1c, lipids, weight
- Treat aggressively; statins are safe with CNI (watch interactions)



Endocrine, fertility and sexual health

Problem	Cause	Management
Hypothyroidism	TBI, neck irradiation, busulfan	Yearly TSH; levothyroxine
Adrenal insufficiency	Steroid withdrawal	Slow taper; stress dosing during illness
Growth hormone deficiency (children)	TBI, cranial RT	Endocrine review; GH replacement
Hypogonadism	Alkylators, TBI; almost universal after MAC in women	Hormone replacement; oestrogen for bone and symptoms; testosterone if low
Osteoporosis	Steroids, hypogonadism, CNI	Ca/vitamin D, bisphosphonates, DXA
Sexual dysfunction	Hormonal, genital cGVHD, psychological	Ask actively; topical oestrogen, dilators, counselling



PART VII · RELAPSE AND CELLULAR THERAPY

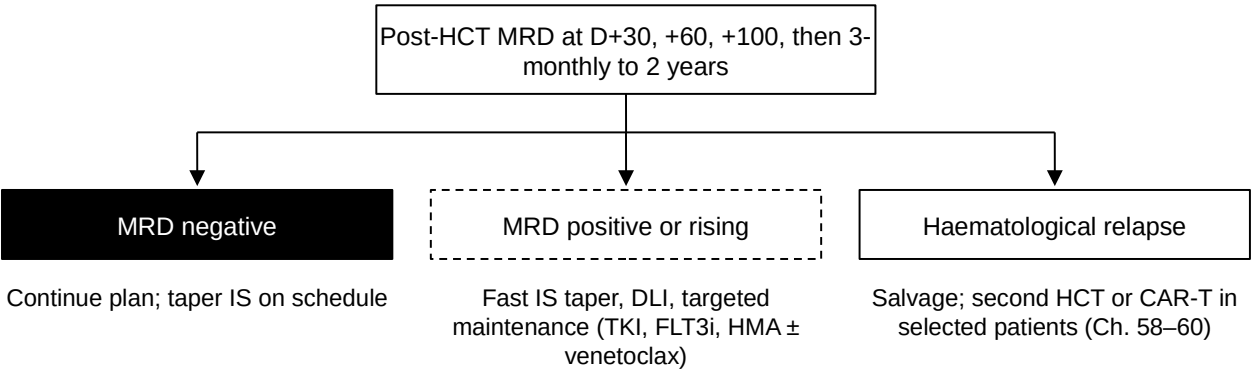
Relapse Prevention, Cellular and Gene Therapy

MRD monitoring, pharmacological maintenance, donor lymphocyte infusion, CAR-T cells, immune escape and ATMP regulation.

Chapters 57–62

MRD monitoring in ALL and AML

Method	Sensitivity	Use
Multiparameter flow cytometry	10 ⁻⁴	ALL and AML (LAIP / different-from-normal)
qPCR of IG/TR rearrangements	10 ⁻⁵	ALL standard (EuroMRD)
qPCR of fusion / mutation	10 ⁻⁵ –10 ⁻⁶	BCR::ABL1, NPM1, RUNX1::RUNX1T1, CBFB::MYH11, WT1 expression
NGS	10 ⁻⁶	IG/TR in ALL; mutation panels in AML (exclude DTA clones)



Pre-HCT MRD positivity is the strongest predictor of relapse in both AML and ALL and should drive conditioning and maintenance decisions, not exclude transplant.

Pharmacological prevention and treatment of relapse

Class	Drugs	Disease / evidence
BCR::ABL1 TKI	Imatinib, dasatinib (CNS penetration), ponatinib (T315I)	Ph+ ALL: restart after engraftment, prophylactic or MRD-triggered; CML blast crisis
FLT3 inhibitors	Sorafenib, midostaurin, gilteritinib	FLT3-ITD AML: sorafenib maintenance improved RFS (SORMAIN); gilteritinib benefit in MRD+ (MORPHO)
Hypomethylating agents	Azacitidine, decitabine	AML/MDS: pre-emptive at MRD relapse (RELAZA2); oral azacitidine maintenance under study
BCL-2 inhibitor	Venetoclax	Relapsed AML with HMA or low-dose cytarabine
Checkpoint inhibitors	Nivolumab, pembrolizumab	Hodgkin lymphoma; high GVHD risk after allo-HCT — low dose, caution
IDH / menin inhibitors, antibodies	Ivosidenib, enasidenib, revumenib; blinatumomab, inotuzumab	Targeted salvage bridging to DLI or second HCT

- Maintenance drugs act best on low disease burden and may add GVL through immunomodulation (HMA → Tregs, sorafenib → IL-15).
- Start after engraftment and count recovery; watch for cytopenia and interactions with CNI (azoles!).

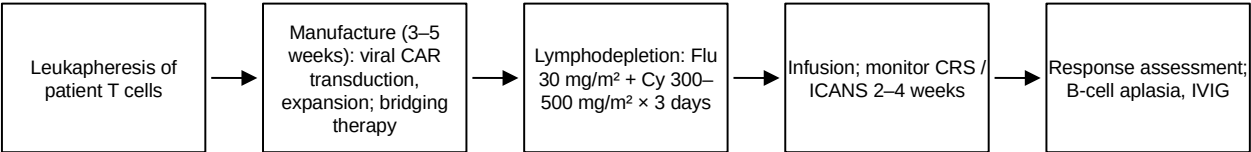
Donor lymphocyte infusion

Intent	Trigger	Comment
Prophylactic	High-risk disease, MRD-negative, ≥ D+100–120, no GVHD, off IS	Escalating doses; best evidence in AML / MDS
Pre-emptive	MRD positive or falling donor chimerism	Stop IS first; combine with HMA or TKI
Therapeutic	Overt relapse	Cytoreduce first; response best in CML, poor in ALL

Setting	Starting CD3+ dose / kg	Escalation
Matched sibling	$1 \times 10^6 - 1 \times 10^7$	Half-log to 1-log every 4–8 weeks
Unrelated donor	$1 \times 10^6 - 5 \times 10^6$	As above
Haploidentical	$1 \times 10^5 - 1 \times 10^6$	Very cautious
CML (therapeutic)	1×10^7	Up to 10^8

- Contraindications: active GVHD, ongoing high-dose immunosuppression, uncontrolled infection.
- Toxicity: GVHD in 30–50% (the price of GVL), marrow aplasia if recipient haematopoiesis dominant; GVHD after DLI predicts response.

CAR-T cells and gene therapy



Product	Target	Approved indications (2024)
Tisagenlecleucel (Kymriah)	CD19	B-ALL ≤ 25 y R/R, DLBCL ≥ 3rd line, FL
Axicabtagene ciloleucel (Yescarta)	CD19	LBCL 2nd line (ZUMA-7) and ≥ 3rd line, FL
Lisocabtagene maraleucel (Breyanzi)	CD19	LBCL 2nd line (TRANSFORM), FL, MCL, CLL
Brexucabtagene autoleucel (Tecartus)	CD19	MCL, adult B-ALL
Idecabtagene vicleucel (Abecma)	BCMA	Myeloma (2nd line and later after KarMMa-3)
Ciltacabtagene autoleucel (Carvykti)	BCMA	Myeloma (2nd line and later after CARTITUDE-4)

CRS (ASTCT grading)

- G1 fever only; G2 hypotension not needing pressors and/or O₂ ≤ 6 L/min; G3 one pressor or high-flow O₂; G4 multiple pressors or ventilation
- Tocilizumab 8 mg/kg (max 800 mg, up to 3–4 doses) ± dexamethasone from grade 2

ICANS

- ICE score (orientation, naming, commands, writing, attention) plus consciousness, seizures, motor, oedema
- Dexamethasone 10 mg q6h from grade 2; high-dose methylprednisolone for grade 3–4; anakinra emerging

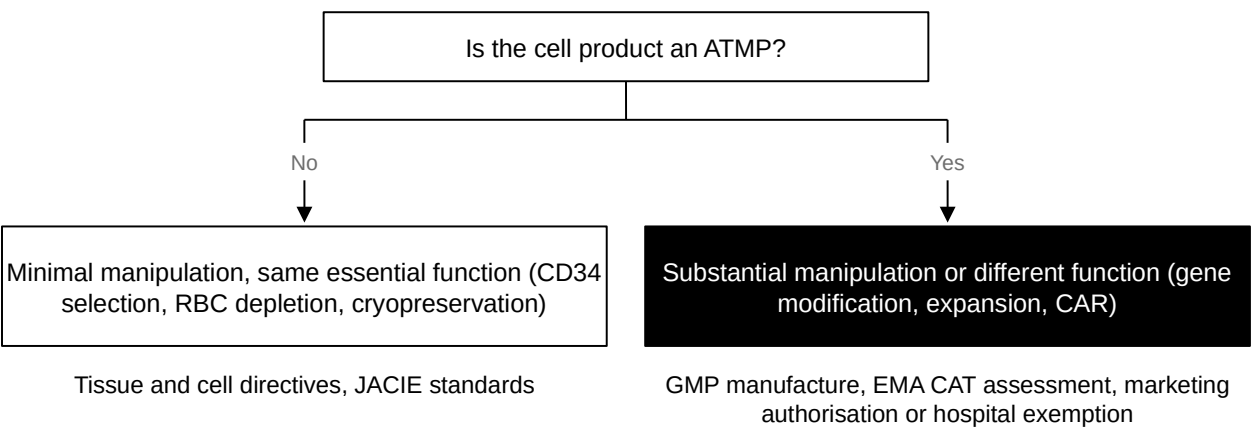
Mechanisms of immune resistance

Escape mechanism	How it works	Consequence
HLA loss (uniparental disomy 6p)	Leukaemia deletes the mismatched haplotype after haplo-HCT	DLI from the same donor is useless; different donor needed
HLA class II downregulation	Loss of antigen presentation to CD4 T cells	Relapse after matched HCT; IFN- γ or HMA may restore
Checkpoint upregulation	PD-L1, CTLA-4 ligands on blasts; exhausted T cells	Checkpoint blockade (GVHD risk)
Antigen loss / lineage switch	CD19-negative relapse after blinatumomab or CAR-T; myeloid switch in KMT2A ALL	Switch to CD22 or dual-targeting CAR-T
Suppressive microenvironment	Tregs, MDSC, IDO, TGF- β	Poor GVL despite full chimerism

- Practical: at relapse after haplo-HCT, test the leukaemia for HLA loss before offering DLI.
- Combining immunomodulation (HMA, checkpoint) with cellular therapy is an active research area.

Regulatory aspects: ATMPs

ATMP class (EU Regulation 1394/2007)	Examples
Gene therapy medicinal products	Lentiviral HSC gene therapy, CRISPR-edited HSC (exa-cel)
Somatic cell therapy medicinal products	CAR-T cells, virus-specific T cells, MSC
Tissue-engineered products	Engineered skin, cartilage
Combined ATMPs	Cells on a scaffold or device



Hospital exemption allows non-routine, in-house ATMPs for individual patients under national authority oversight. Costs and centre qualification requirements are major access barriers.

PART VIII · SPECIFIC MODALITIES

Specific Modalities of HCT

At-home transplantation, cord blood, haploidentical HCT, extracorporeal photopheresis, obese and elderly patients, and HCT in resource-limited settings.

Chapters 63–69

At-home autologous HCT

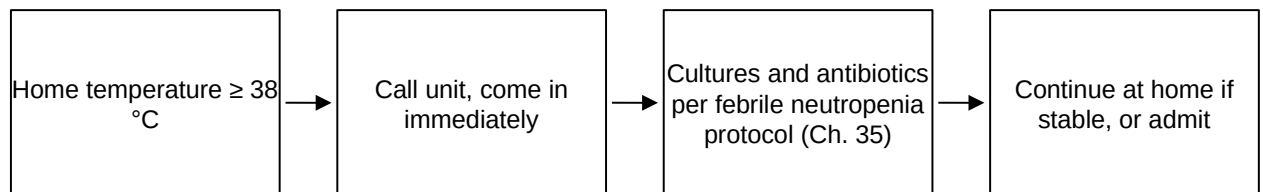
- Concept: after conditioning and infusion the patient spends the aplastic phase at home (or hotel) with daily nurse visits or telemonitoring instead of on the ward.
- Benefits: fewer nosocomial infections and less *C. difficile*, better quality of life, lower cost, freed beds; similar safety in selected patients.

Eligibility

- KPS $\geq$ 80%, no severe organ dysfunction
- 24-h caregiver at home
- Home within $\sim$ 1 h of the centre
- Trained team, 24-h phone line, rapid readmission path

Readmission causes (by frequency)

- Fever / infection ($\approx$ 50%)
- Mucositis with inability to eat or drink
- Pain control
- Need for intravenous hydration or transfusion



Umbilical cord blood transplantation

Advantages

- Rapid availability (weeks), no donor risk
- Tolerates 1–2 HLA mismatches (4–6/6)
- Less chronic GVHD despite mismatch
- Strong GVL; low relapse in MRD+ leukaemia (Milano 2016)

Disadvantages

- Slow engraftment (neutrophils median D+25–28, platelets D+40–60)
- Higher graft failure and early infection
- Slow immune reconstitution; HHV-6, adenovirus
- Limited cell dose for adults; cost

Selection rule	Threshold
Cell dose (single unit)	TNC $\geq 3 \times 10^7$ /kg, CD34+ $\geq 1.5 \times 10^5$ /kg recipient
HLA	$\geq 4/6$ antigen-level; prefer $\geq 5/8$ allele-level; avoid donor-specific antibodies
Double cord	Adults when no single unit reaches dose; one unit finally dominates
Expansion products	Omidubicel (UM171, nicotinamide-expanded) shortens neutrophil recovery

Use of cord blood has fallen as haplo-PTCy grew; it remains valuable for rare HLA types, urgent transplants and children with non-malignant diseases.

Haploidentical HCT

Platform	Principle	Where used
PTCy (Baltimore)	Cyclophosphamide 50 mg/kg D+3, +4 kills alloreactive T cells; then tacrolimus + MMF	Adults worldwide; BM or PBSC
GIAC / Beijing	G-CSF-primed BM + PBSC, ATG, intensive IS	China
TCRαβ/CD19 depletion	Ex vivo removal of αβ T and B cells	Children (Italy, Germany)
CD34 selection ('megadose')	Highly purified stem cells	Historical (Perugia)

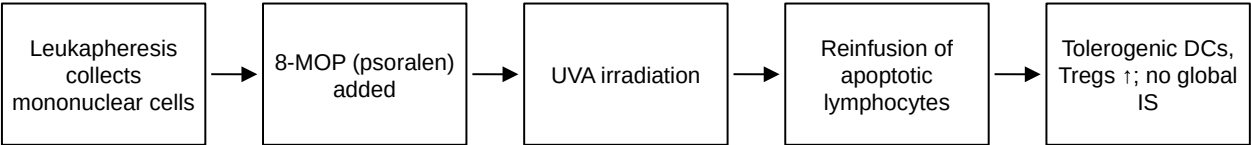
Outcomes with PTCy

- aGVHD grade III–IV ≈ 10–15%, cGVHD moderate–severe ≈ 15–20%
- Survival comparable to matched unrelated donors in registry studies
- Slower CD4 recovery → more CMV: letermovir to D+100 or beyond

Donor choice

- Younger donor; avoid donor-specific anti-HLA antibodies (test MFI)
- Avoid mother and female-to-male if alternatives
- CMV-seropositive donor if recipient is positive
- KIR B haplotype / NK alloreactive donor in AML if available

Extracorporeal photopheresis



Indication	Response	Typical schedule
Steroid-refractory aGVHD (skin, liver, gut)	Skin CR ≈ 60%; liver and gut lower	2 consecutive days weekly, then taper
cGVHD skin (sclerotic), mouth, liver	Best responses; steroid-sparing	2 days every 2 weeks for 3 months, then monthly
Lung cGVHD (BOS)	Stabilisation rather than improvement	As above

- Advantages: very low toxicity, no increase in infection or relapse, combinable with drugs.
- Limits: venous access, time, cost; slow onset (allow 8–12 weeks before judging).

HCT in overweight and obese patients

- Obesity changes drug distribution, protein binding and clearance; body-weight-based dosing can over- or under-dose.
- Obesity itself (BMI > 35 scores 1 in HCT-CI) modestly increases NRM; do not deny HCT on BMI alone.

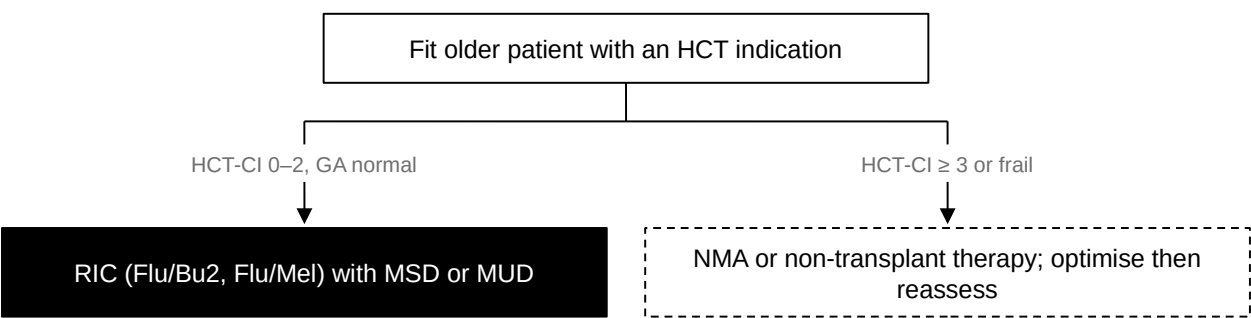
Drug	Dosing weight	Note
Busulfan	Adjusted body weight (AIBW); TDM	Target AUC; highly variable
Cyclophosphamide	Adjusted body weight	$ABW = IBW + 0.4 \times (actual - IBW)$
Fludarabine	Actual weight via BSA; cap BSA $\approx 2.0\text{ m}^2$ or adjust for renal function	Neurotoxicity if overdosed
Melphalan	BSA (some cap at 2.0 m^2)	Mucositis
ATG, G-CSF	Actual weight (G-CSF); ATG per protocol	—
Treosulfan, thiotepa	BSA	—

IBW (Devine): men 50 kg, women 45.5 kg, plus 2.3 kg per inch over 152 cm. $ABW = IBW + 0.4 \times (actual - IBW)$. Calculator: `hct_tools.py`.

HCT in elderly patients

- Age ≥ 60–65 years; now > 30% of allo-HCT in Europe thanks to RIC/NMA. Chronological age alone is not a contraindication.
- Main hazards: infection with slower immune recovery, poor tolerance of cGVHD, frailty and polypharmacy.

Tool	What it captures	Cut-off
HCT-CI (± age)	Comorbidity burden	≥ 3 = high risk
KPS	Functional status	≥ 70% expected
Geriatric assessment	IADL, cognition, gait speed, falls, nutrition, mood, polypharmacy	Any impairment → prehabilitation, RIC/NMA
EBMT / disease risk index	Disease-related risk	Balance against NRM



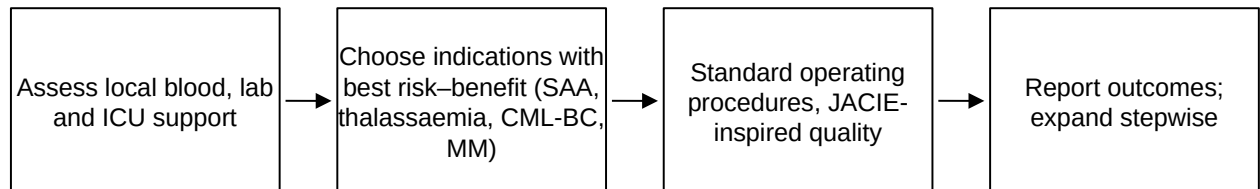
HCT in resource-limited settings

Constraints

- Blood bank capacity (platelets), irradiators
- HLA typing and CMV/EBV PCR cost
- Cryopreservation, GMP processing
- Trained staff, ICU access, drug availability (letermovir, posaconazole, defibrotide)

Strategies

- Start with autologous HCT for myeloma and lymphoma
- Allo-HCT first with matched siblings; then haplo-PTCy (no registry costs)
- Fresh (non-cryopreserved) grafts, oral busulfan, cheaper prophylaxis (fluconazole, aciclovir)
- Twinning with established centres; WBMT and EBMT training; registry participation



PART IX · INDICATIONS AND RESULTS

Indications and Results by Disease

Acute leukaemias, MDS and MPN, marrow failure, haemoglobinopathies, plasma cell disorders, lymphomas, immune deficiencies, autoimmune and metabolic diseases, solid tumours and CAR-T indications.

Chapters 70–95

AML in adults

ELN 2022 risk	Allo-HCT in CR1	Rationale
Favourable (CBF, NPM1 without adverse, bZIP CEBPA)	Not routinely	Chemotherapy cures > 50%; transplant at MRD persistence or relapse
Intermediate	Yes, if donor available	Relapse ≈ 50% with chemo alone; MRD status refines
Adverse (TP53, complex or monosomal karyotype, ASXL1, RUNX1, FLT3-ITD high)	Strongly	Chemotherapy rarely curative; proceed at first CR
MRD positive after induction / consolidation (any risk)	Yes	MRD trumps cytogenetic category

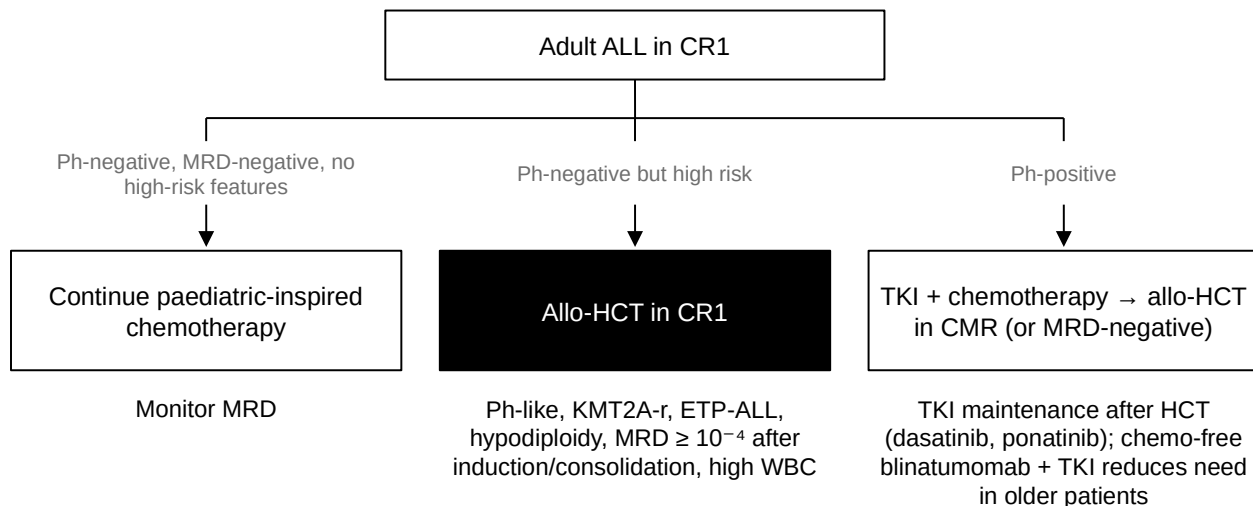
- Beyond CR1: allo-HCT still cures a third in CR2; active disease transplants have high NRM and relapse but sequential (FLAMSA-like) approaches exist.
- Conditioning: MAC for fit patients < 55–60 y (less relapse, BMT CTN 0901); RIC for older or comorbid; MRD+ may favour MAC or maintenance.
- Maintenance: FLT3 inhibitors for FLT3-ITD, HMA-based in trials (Ch. 58).
- **APL**: ATRA + arsenic cures without HCT; auto-HCT in second molecular remission, allo if MRD persists.

AML in children

Risk group	Features	HCT in CR1
Standard / low	t(8;21), inv(16), NPM1, bZIP CEBPA, good MRD response	No; HCT at relapse (CR2)
Intermediate	None of the others; MRD-guided	Chemotherapy ± HCT if MSD
High	FLT3-ITD with high allelic ratio, KMT2A fusions other than t(9;11), monosomy 7, del(5q), TP53, 12p abnormalities, MRD > 0.1% after induction 2	Yes, best donor available

- Conditioning: **busulfan-based** (Bu/Cy ± melphalan, Bu/Flu) preferred; avoid TBI for growth and endocrine late effects; treosulfan gaining ground.
- Infants and children < 2 y: busulfan dosed by weight bands with TDM; AUC targeting reduces SOS/VOD.
- FLT3-ITD: sorafenib maintenance after HCT (AAML1031 experience); MRD+ post-HCT → DLI with caution.
- Outcomes: 5-year OS after HCT in CR1 ≈ 60–70%; relapse remains the main failure.

ALL in adults



- Conditioning: TBI-based MAC (Cy/TBI 12 Gy or Flu/TBI) gives less relapse than chemo-only in ALL; RIC for > 55 y.
- Relapsed / refractory: blinatumomab or inotuzumab to MRD-negative CR, then HCT; CD19 CAR-T (brexu-cel, obe-cel) as alternative or bridge.
- Post-HCT: MRD monitoring with IG/TR PCR; MRD+ → IS taper, DLI, blinatumomab or TKI.

ALL in children and adolescents

HCT indications (CR1)

- Induction failure; poor early MRD response ($\geq 10^{-3}$ after consolidation on ALL-IC/AIEOP-BFM lines)
- Ph+ with poor MRD response; KMT2A::AFF1 infant ALL with high risk; hypodiploidy < 44 chromosomes
- Persistent MRD before HCT is the strongest adverse factor — clear it (blinatumomab, CAR-T)

Element	Recommendation (FORUM trial)
Donor	MSD $\approx$ MUD (10/10 or 9/10); haplo (TCR $\alpha\beta$ depletion or PTCy) if none; cord blood acceptable
Source	Bone marrow preferred (less cGVHD); PBSC for haplo depletion platforms
Conditioning $\geq$ 4 years	TBI 12 Gy + etoposide is superior to chemo-conditioning (FORUM): fewer relapses, better OS
Conditioning < 4 years	Chemotherapy: Flu/Thio/Bu or Flu/Thio/Treo
Post-HCT	MRD and chimerism at D+30, +60, +100, +180, then 3-monthly; pre-emptive IS taper or DLI

MDS in adults

IPSS-R (score)	IPSS-M	Recommendation
Very low (≤ 1.5) / Low ($> 1.5-3$)	Very low / Low / Moderate-low	Observe or supportive care; HCT delayed until progression (survival gain from waiting)
Intermediate ($> 3-4.5$)	Moderate-high	Individualise: age, comorbidity, transfusion need, mutations (TP53, ASXL1, RUNX1)
High ($> 4.5-6$) / Very high (> 6)	High / Very high	Allo-HCT as soon as feasible — only curative option (BMT CTN 1102 survival benefit)

- Pre-HCT therapy: HMA or intensive chemotherapy to reduce blasts $\geq 10\%$; complete remission is not required and delay carries risk.
- Conditioning: MAC or RIC give similar OS overall; MAC lowers relapse in fit younger patients; treosulfan-based (Flu/Treo) is a well-tolerated middle ground.
- TP53-mutated MDS: poor outcome regardless; HCT still offers the only long-term survival in a minority.
- Post-HCT relapse prevention: azacitidine-based maintenance or pre-emptive therapy on MRD/chimerism (Ch. 58).

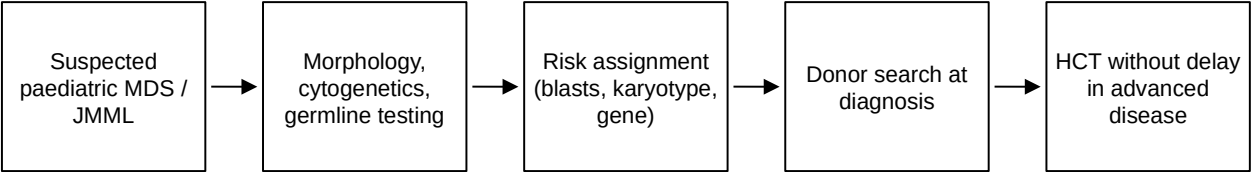
MDS and JMML in children

Refractory cytopenia of childhood / advanced MDS

- Allo-HCT is the only curative therapy; screen for germline predisposition (GATA2, SAMD9/9L, RUNX1)
- RCC with hypocellular marrow and normal karyotype: IST or watchful waiting possible; monosomy 7 → HCT
- Conditioning: Flu/Thio/Treo or Bu-based; avoid TBI; MSD or MUD preferred, haplo feasible

JMML

- RAS-pathway driven (PTPN11, NRAS, KRAS, CBL, NF1)
- Allo-HCT is the only cure for most; relapse 25–35% → early IS taper, second HCT works
- Bu/Cy/Mel or Bu/Flu conditioning; azacitidine bridging improves disease control
- CBL-germline and some NRAS cases may regress spontaneously — observe



MDS/MPN overlap syndromes

Entity	Key features	HCT role
CMML	Monocytosis $\geq 0.5 \times 10^9/L$ and $\geq 10\%$ of WBC; ASXL1, TET2, SRSF2; CPSS-Mol risk	Allo-HCT for intermediate-2 / high CPSS-Mol in fit patients; HMA bridging optional
Atypical CML (BCR::ABL1-negative)	Dysplastic neutrophilia, SETBP1, ETNK1	Poor prognosis → HCT if fit
MDS/MPN with ring sideroblasts and thrombocytosis	SF3B1 + JAK2	Indolent; HCT rarely
MDS/MPN unclassifiable	Mixed features	Individualised

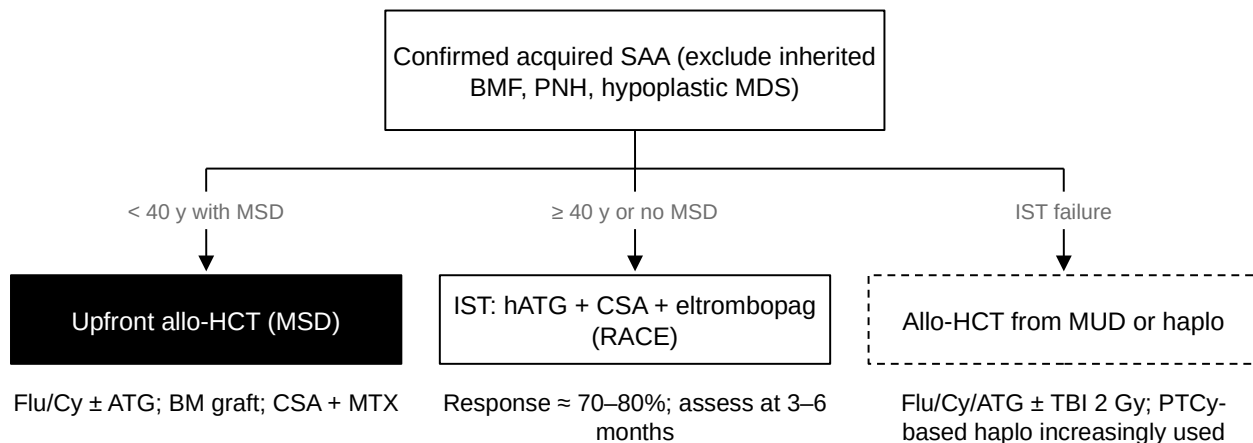
- Mostly older patients → RIC (Flu/Bu, Flu/Mel, Flu/Treo).
- Relapse is the main problem; spleen size and high blast count predict worse outcome; consider ruxolitinib or HMA pre-HCT for proliferative disease.

Myeloproliferative neoplasms

Myelofibrosis risk (DIPSS-plus, MIPSS70)	HCT recommendation
Intermediate-2 or high	Allo-HCT recommended in eligible patients (< 70 y)
Intermediate-1	Consider if transfusion-dependent, high-risk mutations (ASXL1, EZH2, SRSF2, IDH, U2AF1), triple-negative or > 2% blasts
Low	Not recommended

- Ruxolitinib bridging shrinks spleen and improves performance; continue until conditioning and taper to avoid rebound.
- Conditioning: Flu/Bu (RIC or MAC dose), Flu/Mel or Flu/Treo; splenectomy rarely, only for massive spleen with poor graft function history.
- Poor graft function and slow engraftment are common; monitor chimerism and driver-mutation MRD; DLI for molecular relapse works well.
- **CML:** HCT only for blast crisis (after TKI-induced second chronic phase), failure or intolerance of ≥ 2 TKIs including ponatinib/asciminib, or T315I with no options; results best in chronic phase.

Severe aplastic anaemia



- Upfront MUD HCT is a reasonable option in children and young adults when a 10/10 donor is quickly available.
- Avoid transfusion-related sensitisation: leucodepleted, irradiated products; minimise family-member donations.
- Graft rejection is the key risk: BM graft, ATG in conditioning, adequate CSA levels for ≥ 12 months.
- Late clonal evolution (PNH, MDS) after IST — monitor CBC and marrow.

Fanconi anaemia and inherited BMF syndromes

Syndrome	Defect	HCT indication	Special conditioning / issues
Fanconi anaemia	DNA repair; chemo/radio hypersensitivity	Transfusion-dependent BMF, clonal evolution, MDS/AML	Low-dose Flu/Cy (± ATG, ± TBI ≤ 2 Gy); never standard MAC; lifelong head-and-neck SCC screening
Dyskeratosis congenita / telomeropathies	Telomere maintenance	BMF; not the lung or liver disease	RIC (Flu/Cy/alemtuzumab); pulmonary and hepatic fibrosis progress despite HCT
Diamond-Blackfan anaemia	Ribosomal	Steroid-refractory transfusion dependence; MDS	Flu/Bu or Flu/Treo RIC; OS > 85% with MSD/MUD
Shwachman-Diamond	Ribosomal (SBDS)	Severe cytopenia, MDS/AML	RIC; higher cardiotoxicity with high-dose Cy
Severe congenital neutropenia	ELANE, HAX1	G-CSF-refractory, MDS/AML	Standard conditioning

Chromosomal breakage (DEB/MMC) testing and telomere length should precede HCT in any young patient with BMF; siblings must be tested before donating.

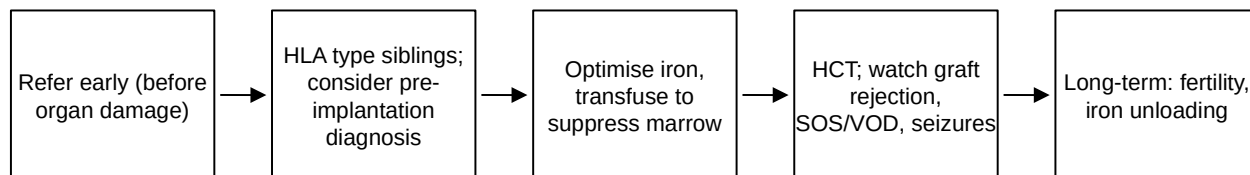
Haemoglobinopathies

Sickle cell disease

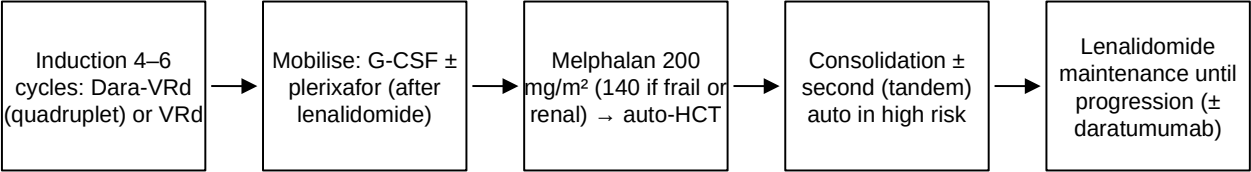
- Indications: stroke or silent infarcts, recurrent acute chest syndrome or vaso-occlusive crises despite hydroxyurea, alloimmunisation with transfusion need
- MSD HCT in children: EFS > 90%; Bu/Cy/ATG or Flu/Bu/ATG; conditioning-related seizures, PRES → keep platelets > 50, Hb 9–11, magnesium
- No MSD: haplo-PTCy with low-dose TBI or gene therapy (exa-cel, lovo-cel)

Thalassaemia major

- Pesaro class I–II: disease-free survival > 90% with MSD; class III (hepatomegaly, fibrosis, poor chelation) 75–85% with pre-HCT cytoreduction
- Age < 14 y and good chelation predict best outcome
- Bu/Cy/Thio or Flu/Bu/Thio; mixed chimerism > 25–30% donor is enough for transfusion independence
- Gene therapy (beti-cel, exa-cel) for those without MSD



Multiple myeloma



Question	Current answer
Who is transplant-eligible?	Fitness rather than age; usually ≤ 70–75 y with adequate organ function
Upfront vs delayed auto?	Upfront still improves PFS (IFM 2009, DETERMINATION); delayed acceptable in standard risk with MRD-negativity — debated
Tandem auto?	Consider for high-risk cytogenetics (del17p, t(4;14), t(14;16)) — EMN02/HO95
Allo-HCT?	Not standard; trials or young high-risk patients with early relapse; NRM 10–20%
CAR-T / bispecifics?	Ide-cel and cilta-cel from second line (KarMMa-3, CARTITUDE-4); teclistamab, elranatamab, talquetamab in relapsed disease; may reshape auto-HCT's role

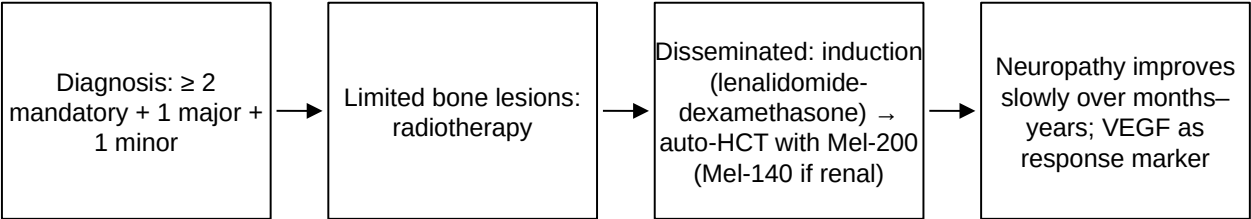
Systemic light-chain (AL) amyloidosis

Eligibility for auto-HCT	Criterion
Cardiac	Mayo stage I–II (NT-proBNP < 5000 ng/L, troponin T < 0.06 µg/L), NYHA ≤ II, LVEF > 40–45%, systolic BP ≥ 90
Renal	eGFR ≥ 30–50 mL/min (dialysis patients in selected centres)
Performance	ECOG ≤ 2, age ≤ 70, ≤ 2 organs severely involved
Other	No significant autonomic neuropathy, GI bleeding or pleural effusions

- Only ≈ 20–25% of patients qualify; strict selection has cut treatment-related mortality from > 20% to < 5%.
- Melphalan 200 mg/m² for fit patients, 140 mg/m² risk-adapted (age, renal function).
- Daratumumab-CyBorD (ANDROMEDA) is now standard induction and a strong non-transplant alternative; the incremental value of auto-HCT after Dara-CyBorD is under study.
- Aim: haematological CR/VGPR; consolidate with daratumumab or bortezomib-based therapy if < VGPR.

POEMS syndrome

Letter	Feature	Status in diagnosis
P	Polyneuropathy (demyelinating, symmetric)	Mandatory
O	Organomegaly, Castleman disease	Minor
E	Endocrinopathy (not diabetes or thyroid alone)	Minor
M	Monoclonal plasma cell disorder, almost always λ	Mandatory
S	Skin changes; also sclerotic bone lesions, papilloedema, oedema, thrombocytosis; VEGF $\uparrow$	Major (sclerotic lesions, Castleman, VEGF) / minor



Outcomes: haematological and VEGF response > 80%, 5-year OS > 90%. Engraftment syndrome is frequent — pre-emptive steroids around neutrophil recovery.

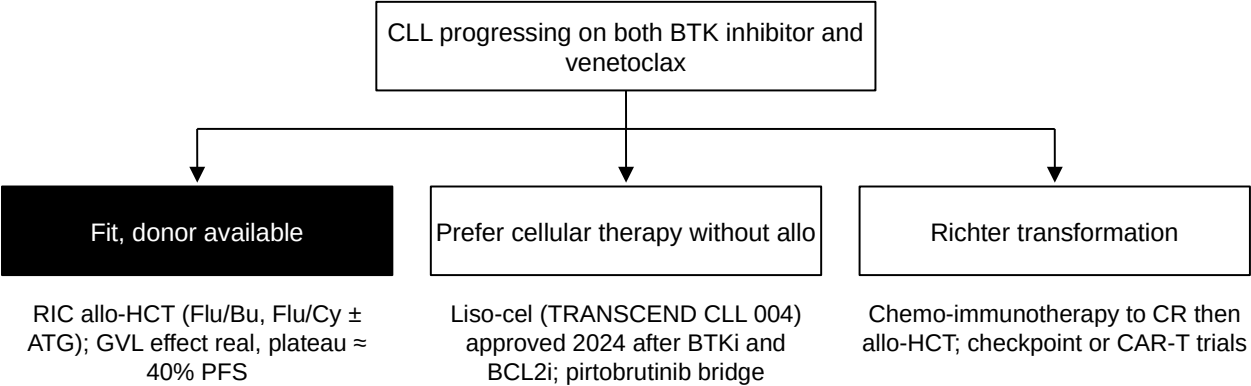
Indolent lymphomas

Setting (follicular and other iNHL)	Option	Comment
Chemosensitive relapse, esp. POD24	Auto-HCT	Prolongs PFS; used less since bispecifics and CAR-T; still relevant for early relapse after chemo-immunotherapy
Multiple relapses, relapse after auto, transformation	Allo-HCT (RIC)	Strong GVL, plateau in survival; NRM 15–25%
≥ 3rd line FL	CD19 CAR-T (axi-cel ZUMA-5, tisa-cel ELARA, liso-cel)	CR ≈ 70–80%; durable in most
Relapsed FL	Bispecifics (mosunetuzumab, epcoritamab), R ² (lenalidomide-rituximab)	Off-the-shelf alternatives

- Marginal zone and lymphoplasmacytic lymphoma: transplant only for multiply relapsed or transformed disease.
- Rituximab maintenance after auto-HCT improves PFS in FL.

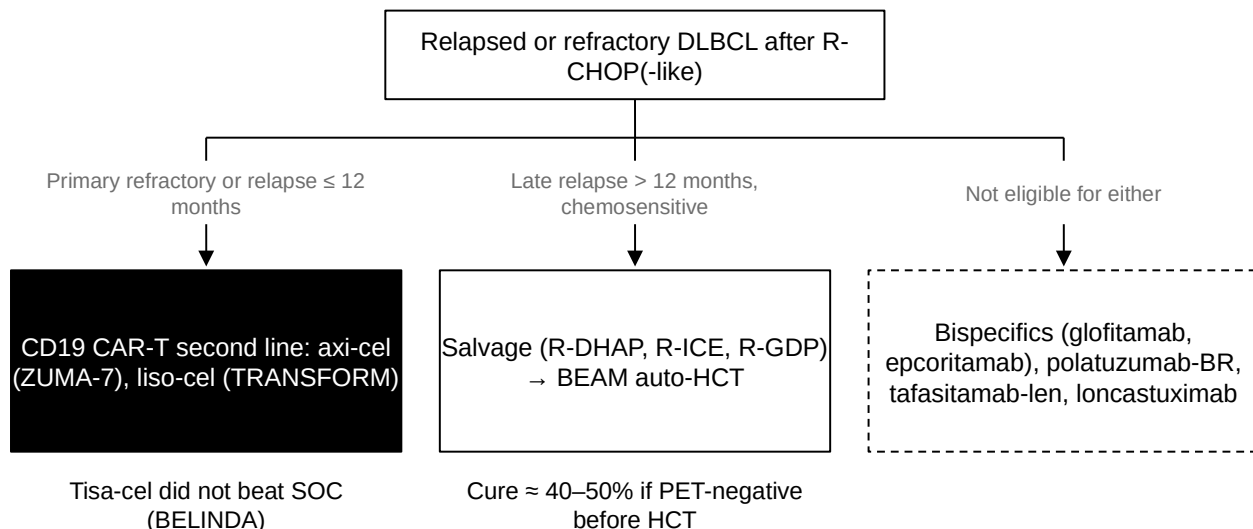
Chronic lymphocytic leukaemia

- Targeted agents (BTK inhibitors, venetoclax, ± obinutuzumab) have made allo-HCT a late option; auto-HCT has no role.



Discuss allo-HCT before double-refractory disease develops; TP53 aberration alone no longer mandates transplant.

Diffuse large B-cell lymphoma



- CAR-T ≥ 3rd line: axi-cel (ZUMA-1), tisa-cel (JULIET), liso-cel (TRANSCEND) — durable CR ≈ 35–40%.
- Allo-HCT: relapse after auto-HCT or CAR-T in fit patients; RIC; NRM 20–30%, long-term PFS ≈ 30%.
- Auto-HCT in CR1 is not recommended outside trials, even for high IPI or double-hit lymphoma.

Mantle cell lymphoma

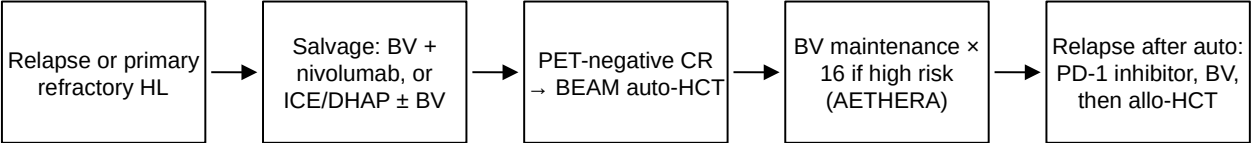
Line	Younger fit patients	Comment
First	Cytarabine-containing induction (R-CHOP/R-DHAP, Nordic) + rituximab maintenance; auto-HCT consolidation	TRIANGLE: ibrutinib added to induction and maintenance; auto-HCT adds little when ibrutinib is given — practice shifting
Relapse	BTK inhibitor (ibrutinib, acalabrutinib, zanubrutinib); pirtobrutinib after cBTKi	Duration limited, especially TP53-mutated
After BTKi failure	Brexucabtagene autoleucel (ZUMA-2): ORR ≈ 90%, CR ≈ 65%	Allo-HCT (RIC) for fit patients as alternative or after CAR-T

- TP53-mutated or blastoid MCL: poor response to chemo-immunotherapy and auto-HCT; consider early BTKi and cellular therapy.
- Allo-HCT retains GVL in MCL; NRM ≈ 20%; reserve for young patients with high-risk relapse.

Other aggressive lymphomas: T-cell, NK/T-cell, HIV-associated

Entity	Standard approach	Transplant role
ALK+ ALCL	CHOP-based (brentuximab-CHP)	Auto only at relapse
PTCL-NOS, AITL, ALK-ALCL	CHOP/CHOEP or BV-CHP	Auto-HCT consolidation in CR1 (widely used, level II evidence); allo for relapse or CR1 in very high risk
Relapsed PTCL	Salvage (GDP, ICE), brentuximab, pralatrexate, romidepsin	Allo-HCT (RIC) curative in ≈ 40%; auto if chemosensitive and not previously given
Extranodal NK/T-cell (nasal)	Asparaginase-based (SMILE, P-GemOx, DDGP) + radiotherapy	Auto-HCT for advanced stage in CR; allo for relapse
HIV-associated (DLBCL, Burkitt, HL)	Standard chemo-immunotherapy with cART	Auto-HCT with same outcomes as HIV-negative if HIV controlled; watch cART–chemo interactions (ritonavir, cobicistat)

Hodgkin lymphoma



Question	Answer
Best predictor of auto-HCT success	PET-negative status before transplant (Deauville ≤ 3)
Allo-HCT indication	Relapse after auto-HCT in fit patients, ideally in response to PD-1 or BV; RIC; haplo-PTCy works well
Checkpoint inhibitor before allo	Effective bridge but more early severe aGVHD and hepatic SOS; leave ≥ 6 weeks, use PTCy
Checkpoint inhibitor after allo	Severe GVHD risk — only for relapse without alternatives, low dose, off IS
Children	Same principles; avoid TBI; auto-HCT for high-risk relapse

Inborn errors of immunity (primary immunodeficiencies)

Disease	HCT urgency	Approach
SCID	Emergency — transplant before infection (newborn screening)	MSD without conditioning; MUD/haplo with RIC (Flu/Treo or Bu); TCRαβ/CD19 depletion for haplo; OS > 90% if < 3.5 months and infection-free
Wiskott-Aldrich	Early in childhood	MAC/RTC Bu-based; full donor chimerism needed for platelets; gene therapy available
Chronic granulomatous disease	Elective, before organ damage	RIC Flu/Bu/ATG or alemtuzumab; excellent survival
Familial HLH	After disease control	HLH-94/2004 (etoposide, dexamethasone, CSA) or emapalumab → RIC HCT (Flu/Treo/alemtuzumab); mixed chimerism ≥ 20–30% protects
CTLA4, LRBA, IPEX, others	Individualised	RIC; targeted drugs (abatacept, sirolimus) as bridge

- Goal is immune reconstitution, not eradication of malignancy → reduced-toxicity conditioning is standard.
- Pre-HCT: treat infections, immunoglobulin replacement, irradiated CMV-safe blood, avoid live vaccines.

HCT for autoimmune diseases

Disease	Who	Evidence	Regimen
Multiple sclerosis	Relapsing, highly active despite high-efficacy DMT, EDSS ≤ 6.5, age < 50	MIST RCT: 5-y progression 10% vs 75% with DMT	Cy/ATG (auto)
Systemic sclerosis	Early diffuse (< 5 y), lung or heart involvement but no severe pulmonary hypertension	ASTIS and SCOT RCTs: better long-term survival; TRM 3–10%	Cy 200 mg/kg + ATG ± TBI (SCOT)
Crohn's disease	Refractory to biologics, unsuitable for surgery	ASTIC: no remission benefit; selected centres	Cy/ATG
SLE, CIDP, NMOSD, type 1 diabetes	Refractory cases	Case series; trials	Cy/ATG or BEAM

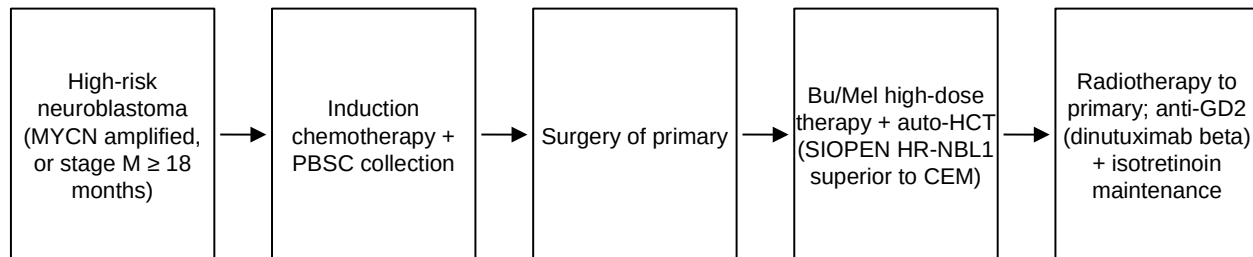
- Mechanism: immune reset — thymic re-education, new Treg and naive T-cell repertoire; autologous only, except rare allo for refractory cases (and CD19 CAR-T now being tested in SLE).
- Accredited centres with joint rheumatology or neurology teams; register in EBMT ADWP.

Inborn errors of metabolism

Disease	HCT indication	Key point
Hurler syndrome (MPS I-H)	Diagnosis < 2–2.5 y, DQ > 70	Standard of care; donor macrophages deliver enzyme; enzyme replacement as bridge
Cerebral X-linked adrenoleukodystrophy	Early cerebral disease (Loes 1–9), no major neurological deficit	MSD/MUD HCT or lentiviral gene therapy (eli-cel); asymptomatic boys need MRI surveillance
Krabbe disease, metachromatic leukodystrophy	Pre-symptomatic (late-infantile) or early juvenile	Gene therapy (atidarsagene) for MLD
Osteopetrosis (malignant infantile)	Early, before vision loss	High SOS/VOD and pulmonary hypertension risk
MPS II, VI, others	Selected	Limited CNS benefit; ERT preferred for many

- Full donor chimerism and normal enzyme levels matter: MAC (busulfan-based with TDM) or Flu/Treo; Bu/Flu/ATG standard.
- HCT halts progression; it does not reverse existing neurological damage — speed of referral determines outcome.

Neuroblastoma



- Tandem auto-HCT (thiotepa/Cy then CEM) improved EFS in COG ANBL0532 and is used in North America; single Bu/Mel is European standard.
- Bu/Mel toxicity: SOS/VOD (up to 20%) → busulfan TDM, defibrotide prophylaxis in some protocols; avoid prior ¹³¹I-MIBG close to HDT.
- Relapse: chemo-immunotherapy (irinotecan/temozolomide + dinutuximab), ¹³¹I-MIBG therapy; allo-HCT experimental.

Other solid tumours

Tumour	Auto-HCT role	Regimen / evidence
Germ cell tumours (relapsed, platinum-refractory)	Established salvage; cures ≈ 50% of second-line, 20–40% later	Tandem high-dose carboplatin/etoposide (TI-CE, Indiana); TIGER trial vs conventional
Medulloblastoma / CNS embryonal tumours in children < 3–5 y	Delays or avoids craniospinal irradiation	Head Start, tandem thiotepa-based HDT
Ewing sarcoma (high risk)	Selected: Bu/Mel consolidation in localised high-risk (Euro-EWING 99 R2Loc)	No benefit in primary metastatic (R3)
Retinoblastoma (metastatic), Wilms (relapsed)	Case series	Thiotepa/carboplatin-based
Breast, ovarian, small-cell lung cancer	No role	Negative randomised trials

CAR-T in haematological malignancies: indications and results

Disease	Products	Pivotal trials	Key result
LBCL ≥ 3rd line	Axi-cel, tisa-cel, liso-cel	ZUMA-1, JULIET, TRANSCEND	CR ≈ 40–55%; ≈ 35–40% long-term
LBCL 2nd line (refractory or relapse ≤ 12 mo)	Axi-cel, liso-cel	ZUMA-7, TRANSFORM	EFS and OS (axi-cel) superior to auto-HCT
FL ≥ 3rd line	Axi-cel, tisa-cel, liso-cel	ZUMA-5, ELARA, TRANSCEND FL	CR ≈ 70–80%
MCL after BTKi	Brexu-cel, liso-cel	ZUMA-2, TRANSCEND MCL	CR ≈ 65%
B-ALL	Tisa-cel (≤ 25 y), brexu-cel (adults), obe-cel	ELIANA, ZUMA-3, FELIX	CR 70–90%; ≈ 50% long-term
Myeloma	Ide-cel, cilta-cel	KarMMa(-3), CARTITUDE-1/-4	PFS gain in 2nd–4th line; cilta-cel CR > 80%
CLL	Liso-cel	TRANSCEND CLL 004	Approved 2024 after BTKi + BCL2i

- CAR-T vs allo-HCT after CAR-T in B-ALL: consolidative allo improves outcome in adults and in children with early B-cell recovery or MRD; not needed for all.
- Relapse after CAR-T: CD19-negative escape, short persistence; options include allo-HCT, bispecifics, alternative-target CAR-T.
- Registry follow-up (EBMT CTIWP) is mandatory: late cytopenias, infections, second malignancies (T-cell lymphoma reports 2024).

Audit notes: corrections (1 of 2)

Every chapter summary in ebmt-handbook/references/*.md was read against the handbook and current guidance while building this deck. These statements were wrong or misleading and are corrected in the slides and in the reference files.

Ch.	Reference file said	Corrected to
11	HCT-CI 0 / 1–2 / ≥3 → NRM ≈15% / 26% / 41%	Sorror 2005: 2-year NRM ≈ 14% / 21% / 41%
16	Poor mobiliser = CD34+ < 10/μL	GITMO: peak CD34+ < 20/μL or < 2 × 10 ⁶ /kg in ≤ 3 aphereses
36	KPC / ESBL → meropenem	ESBL → carbapenem; KPC producers → ceftazidime-avibactam or meropenem-vaborbactam; MBL → aztreonam + ceftazidime-avibactam or cefiderocol
38	Maribavir approved 2022	FDA November 2021 (EMA 2022) for refractory/resistant CMV
43	Ruxolitinib for aGVHD 'FDA 2021'	FDA approval for steroid-refractory aGVHD was 2019 (REACH1); 2021 was cGVHD (REACH3)
49	SOS/VOD: 'any ≥ 2 of' bilirubin, hepatomegaly, ascites, ultrasound	EBMT 2016: bilirubin ≥ 2 mg/dL plus ≥ 2 of painful hepatomegaly, weight gain > 5%, ascites (classical ≤ 21 d); late-onset criteria differ
52	BOS criteria require 'no hyperinflation'	NIH criteria require evidence of air trapping (RV > 120% or expiratory CT) with FEV1/FVC < 0.7 and FEV1 < 75%

Audit notes: corrections (2 of 2)

Ch.	Reference file said	Corrected to
60	CRS grade 2 = 'O ₂ < 40%'	ASTCT: grade 2 = hypotension not needing vasopressors and/or O ₂ by low-flow cannula ≤ 6 L/min
74	IPSS-R low < 3.5, intermediate 3.5–4.5	IPSS-R: very low ≤ 1.5, low > 1.5–3, intermediate > 3–4.5, high > 4.5–6, very high > 6
81	Lenalidomide maintenance 'FORTE/MAIA'	Evidence: CALGB 100104, IFM 2005-02, Myeloma XI meta-analysis; MAIA is a transplant-ineligible trial
82	Mel-140 for normal organ function, Mel-100 if impaired	Mel-200 for fit patients; Mel-140 risk-adapted
86	Axicabtagene ciloleucel labelled 'KymRiah'	Axi-cel is Yescarta; Kymriah is tisagenlecleucel
91	MIST: 'no progression 79% vs 9%'	MIST: progression at 5 years 9.7% (HSCT) vs 75.3% (DMT)

Audit notes: scope and provenance

- The reference files tag each chapter either **verified from the book** or **model-supplemented** (Chinese tags in the source). The footer symbol on each slide carries that tag forward: ● verified, ○ supplemented.
- Supplemented chapters (1–8, 11–25, 27–29, 31–34, 41–42, 45, 61–62, 69, 76, 82–85, 87–94) were rewritten from general knowledge of the 8th edition and current ELN, ECIL, EBMT and ASTCT guidance; treat numbers as teaching values.
- Items added beyond the reference files where the handbook covers them: engraftment definitions (Ch. 21), ABO bidirectional mismatch (Ch. 24), MAGIC aGVHD grading (Ch. 43), NIH cGVHD severity rules (Ch. 44), EASIX (Ch. 42), FORUM trial (Ch. 73), IPSS-M (Ch. 74), TRIANGLE (Ch. 87), FELIX/obe-cel (Ch. 95).
- Newer than the 2024 print edition and flagged as such: 2023 refined SOS/VOD criteria, liso-cel for CLL (2024), KarMMa-3 / CARTITUDE-4 second-line myeloma CAR-T, teclistamab-class bispecifics.
- Not covered by this deck: individual drug monographs, paediatric dosing tables, and the handbook's country-specific regulatory annexes.

Rebuild: node study-materials/src/build.js — content lives in study-materials/src/content/.js; edit and re-run to regenerate PPTX and PDF.*

Abbreviations

Abbr.	Meaning
aGVHD / cGVHD	Acute / chronic graft-versus-host disease
ATG	Anti-thymocyte globulin
ATMP	Advanced therapy medicinal product
BOS	Bronchiolitis obliterans syndrome
Bu / Cy / Flu / Mel / Treo / Thio	Busulfan / cyclophosphamide / fludarabine / melphalan / treosulfan / thioteпа
CNI	Calcineurin inhibitor (ciclosporin, tacrolimus)
CRS / ICANS	Cytokine release syndrome / immune effector cell-associated neurotoxicity syndrome
DLI	Donor lymphocyte infusion
ECP	Extracorporeal photopheresis
GVL	Graft-versus-leukaemia effect
HCT-CI	HCT comorbidity index (Sorrow)
HMA	Hypomethylating agent
IS	Immunosuppression

Abbr.	Meaning
KPS	Karnofsky performance status
MAC / RIC / NMA	Myeloablative / reduced-intensity / non-myeloablative conditioning
MMF / MTX	Mycophenolate mofetil / methotrexate
MRD	Measurable residual disease
MSD / MUD / MMUD	Matched sibling / matched unrelated / mismatched unrelated donor
NRM	Non-relapse mortality
PBSC / BM / CB	Peripheral blood stem cells / bone marrow / cord blood
PRES	Posterior reversible encephalopathy syndrome
PTCy	Post-transplant cyclophosphamide
PTLD	Post-transplant lymphoproliferative disorder
SOS / VOD	Sinusoidal obstruction syndrome / veno-occlusive disease
TA-TMA	Transplant-associated thrombotic microangiopathy
TBI / TDM / TNC	Total body irradiation / therapeutic drug monitoring / total nucleated cells